

THEORY NOTES

ON

Course No:-AGRO-235

Course Title:-Field Crops-I (Kharif Crops)

Credits:-2+1=3

Semester-III (New)

B.Sc(Agri)

Complied & Prepared

By

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Syllabus

1.	Cereals	Rice, Maize, Kharif Sorghum, and Pearl millet
2.	Minor millets	Finger millet, Proso millet, Barn yard millet, Foxtail millet, Little millet, Kodo millet
3.	Pulses	Pigeon pea, Green gram, Black gram, Horse gram, Moth bean, Cowpea
4.	Oilseeds	Groundnut, Sesame, Soybean, Castor, Niger
5.	Fiber crops	Cotton, Jute, Sun hemp, and Dhaincha
6.	Forage crops	Sorghum, Pearl millet, Maize, Cowpea, Cluster bean.
7.	Rainfed Grasses	Rhodes, Marvel, Dinanath, Anjan, Blue penic, Pavana grass and Dongarigavat.
8.	Irrigated Grasses	Gajaraj, Napier, Guinea, Para grass.

Reference Books

1. Hand Book of Agriculture, ICAR Publication, 2006
2. Field Crops- Rajendra Prasad
3. Modern Techniques of Raising Field Crops- Chhidda Singh.
4. Theory and Digest Agronomy- S.S. Cheema
5. Farm productivity, New Century and New Challenges- M.M. Hosmani, B.M. Chittarpur and H.B. Babalad
6. Crop production and Field Experimentation- V.G. Vaidya, K.R. Sahasrabuddhe and V.S. Khuspe
7. Krishidarshani- 2015 by M.P.K.V. Rahuri

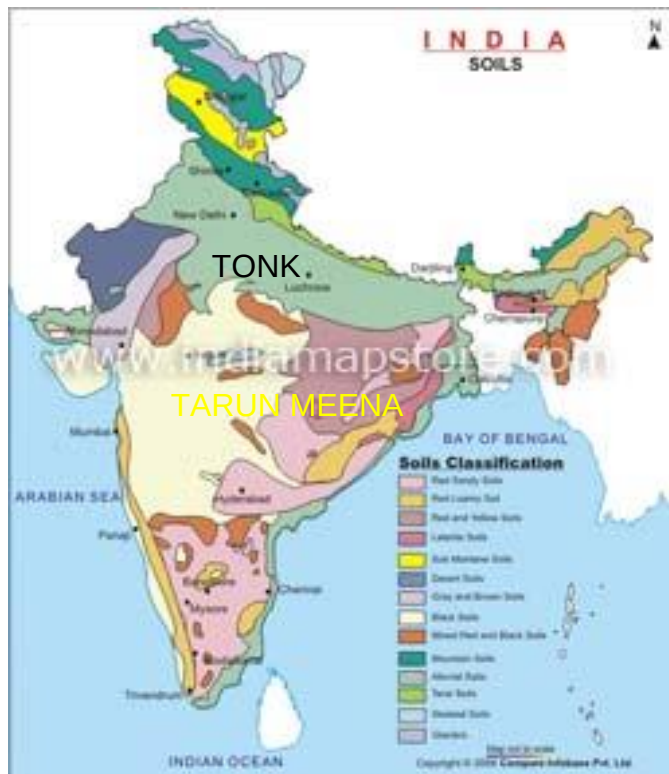
What we are going to study ?

1. Scientific names
2. Economic Importance.
3. Botanical Description.
4. Climatic requirements.
5. Soil requirements.
6. Land preparation.
7. Sowing.
8. Manures and Fertilizer requirements.
9. Water management.
10. Inter-cultivation and Weed Control.
11. Plant protection.
12. Improved Varieties.
13. Harvesting, Threshing and Storage.

Factors governing Crop Production

1. Soil,
2. Water
3. Climate
4. Management

Soils- It act as a medium for plant growth. It is a complex body shows variations in depth, color, composition and behavior. Five things common in all soils: Hard material- mineral matter (45 %), Soft and spongy material- organic matter (5%), Water (25%), Air (25%), Organisms.

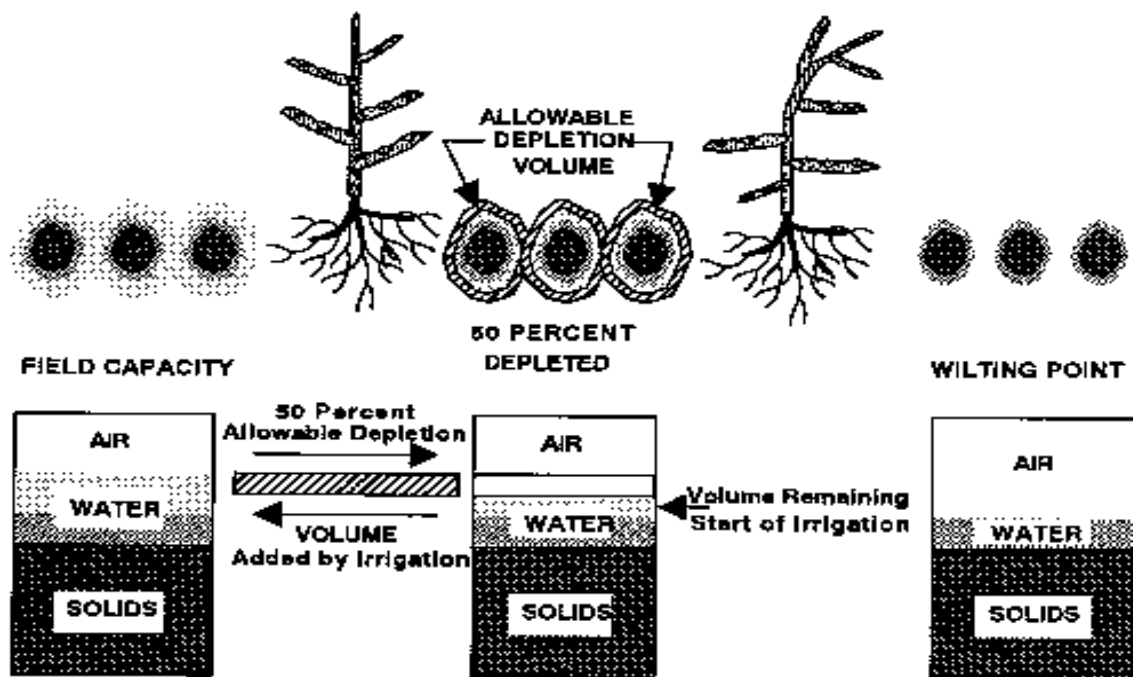


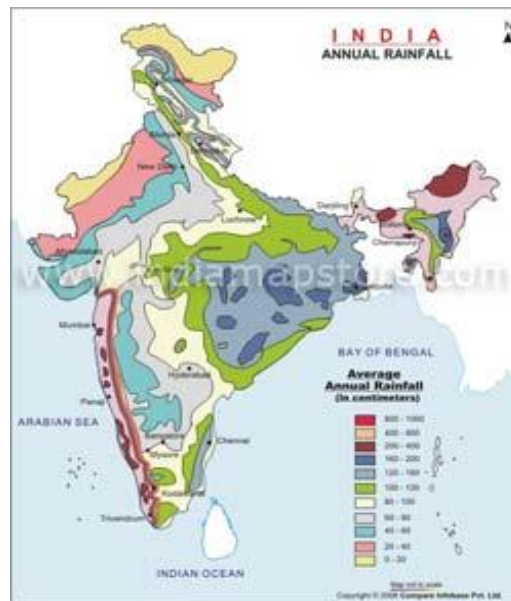
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Major Soil types of India

1. Alluvial soils- Fertile and well drained. Intensive cropping high yields and more profit.
2. Black soils. Deep soils. Swell when wet and become hard when dry.
3. Red soils, Lateritic soils- Developed from lateritic rocks, shallow depth, low silt and clay content.
4. Desert soils.-More sand particles.
5. Hill and valley soils. Sloppy lands, more erodible, need constant conservation.

Water- In India, major source of water is rainfall. Major rainfall occurs through south-west monsoon. By conserving run-off water, the ground water level can be increased. By constructing watersheds, small tanks and big Dams, the stored water can be utilized for irrigation.





Annual Rainfall in India

- 5000 to 10000 mm
- 2500 to 5000 mm
- 1000 to 2500 mm
- 750 to 1000 mm
- Less than 750 mm

Climate- Rainfall (Quantity and Distribution), Temperature (Minimum and Maximum), Humidity are the important weather parameters / factors in crop production.

India- Bio-Diversity - Different types of soils and Climate are prevailing in the country. Hence, different Types of plants in different seasons and animals are there.

Seasons-

Kharif	Rabi	Summer
(June- Sept)	(Oct.- Jan)	(Feb.-May)
Rains	Occasional Rains	no rains
High humidity	Dry climate	Dry climate
Moderate Temp	low temp.	High temp.
Low sunshine	Short day	Long day
Less Evaporation	Moderate Evaporation	High Evaporation

Management- To select proper crop for proper soils and climatic condition. To adopt improved technology for harvesting bumper quality yield.

Agriculture Production in India (m tons)

Crops	2010-11	2011-12	2012-13	2013-14	2014-15
Rice	95.98	105.30	105.24	106.65	103.04
Wheat	86.87	94.88	93.51	95.85	95.76
Sorghum	7.00	5.98	5.28	5.54	4.83
Bajara	10.37	10.28	8.74	9.25	7.91
Maize	21.73	21.76	22.26	24.26	22.97
Small millets	0.44	0.45	0.44	0.43	0.46
Total cereals	226.25	242.20	238.79	245.79	238.64
Tur	2.86	2.65	3.02	3.17	2.75
Gram	8.22	7.70	8.83	9.53	8.28
Green gram	1.80	1.63	1.19	1.61	1.39
Black gram	1.76	1.77	1.90	1.70	1.61
Total pulses	18.24	17.09	18.34	19.78	18.43
Total food grains	244.49	259.29	257.13	265.57	257.07
Crops	2010-11	2011-12	2012-13	2013-14	2014-15
Groundnut	82.65	69.64	46.95	97.14	74.68
Castor	13.50	22.95	19.64	17.27	19.11
Sesamum	8.93	8.10	6.85	7.15	7.01
Niger	1.08	0.98	1.02	0.98	0.99
Mustard	81.79	66.04	80.29	78.77	73.63
Linseed	1.47	1.52	1.49	1.41	1.49
Safflower	1.50	1.45	1.09	1.13	0.78
Sunflower	6.51	5.17	5.44	5.04	4.22
Soybean	127.36	122.14	146.66	118.61	116.41
Total oilseeds	324.79	297.99	309.43	327.49	298.32
Cotton (Lakh bales 170 kg each)	330	352	342.20	359.02	351.52
Sugarcane	3423.82	3610.37	3412.00	3521.42	3549.52

Aims of Crop Production

To produce required quantity of food grains, pulses, oilseeds, sugar, fiber and fodder to feed the nation, and if possible for export by adopting modern scientific techniques in Agriculture.

1. Rice (Paddy): *Oryza sativa* L.

Origin: Rice is regarded as first cultivated crop of Asia. Paddy grains found during excavation at Hastinapur (India) around 1000 – 750 B.C., considered as a oldest sample in the world. De Condolle (1886) mentioned South India is its center of origin. South west Himalaya has various types and varieties, indicated probable center of origin.

Distribution: Rice cultivation is limited as North 49° and south 35°. Most of the world's rice is grown in countries of south and south East Asia, west Africa and Central & south America. China is first in area and production of rice. It is grown in all the states of India.

Area and Production in World

Country	Area (million ha)	Production (Million ton)	Productivity (ton/ha)
China	29.87	137.00	6.55
India	42.86	95.98	3.36
Indonesia	12.08	35.50	4.67
Bangladesh	11.70	31.70	4.06
Thailand	10.67	20.26	2.88
Vietnam	7.60	26.37	5.55
Burma	7.00	10.53	2.35
Philippines	4.53	10.54	3.69
Brazil	2.83	9.30	4.83
Cambodia	2.78	4.23	2.38
World	157.73	449.37	4.25

Economic Importance: Rice is important food crop of India and staple food of 60 % of the world's population. Hulled Rice grain contain 6 - 7 % of proteins, 2 - 2.5 % fat and minerals and Vitamins. By-product of rice milling i.e. husk, bran are used as cattle and poultry feed. Also used in paper industries. Rice straw is good cattle feed and also used in making hats, mats, ropes.

Genus - *Oryza* has 24 species of which 2 are cultivated.

Oryza sativa Var. (2n= 24) found in Asia, America and Europe.

Oryza glaberrima Var. found in West Africa

Indica- Grown in India, Awn less /short and smooth awns. Fruit is caryopsis.

Japonica- Grown in Japan, Grain is oval and round, may be awned or awn less

Javanica- Grown in Indonesia. Stiff straw, long panicle, awned Grains.

Botanical description- Self pollinated Annual Plant, Height- 0.5 to 2.0 m. **Roots**-

Embryonic Adventitious develop from node of Culm. **Shoot**- All plant parts visible above ground- composed of Culm, leaves and panicle. Culm or Stem made of nodes and internodes. Buds near ground

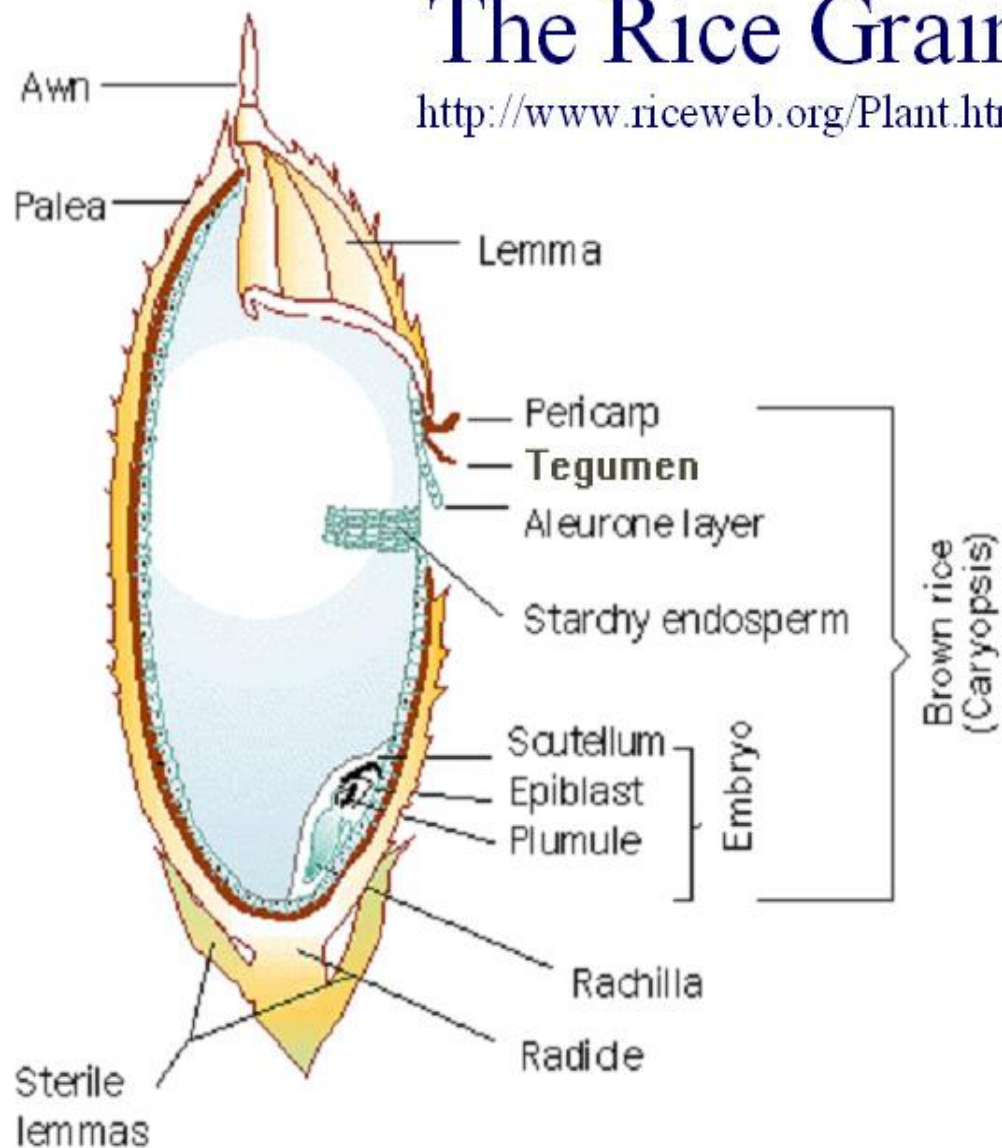
grow in to tillers. Primary tillers gives rise to secondary, Secondary to tertiary tillers. **Leaves**-Sessile born on Culm. **Panicle**- Group of Spike lets. Grain is enclosed by the lemma and palea. The dehulled grain is known as Brown Rice.

Climatic requirements:- Water requirement of rice is 350cm. It needs Warm and Humid Climate (Av. Temp. 21 to 37 °C). At tillering it requires high temp. It is a Short Day plant. Some varieties are non- sensitive to photo period. At panicle initiation it requires 20-22°C temp. At flowering and fertilization- 16- 20°C. At Maturity- 20 - 32°C

Soils Requirement:- Soils with high water retention capacity, with medium clay contain. (clay & clay loam) are ideal soils for rice with sufficient organic matter. It requires pH- 5.5 to 6.5. Semi aquatic plant grows under submerged conditions. Major area is under lowland condition.

The Rice Grain

<http://www.riceweb.org/Plant.htm>



Soils in Maharashtra under Rice Cultivation

Sr. No.	Sub. Region	Soil type	Research Stations
1.	Costal Saline region	Khar lands	Panvel

2.	North konkan region	Non lateritic, medium soils	Karjat, Khopoli , Palghar
3.	South Konkan region	Lateritic	Dapoli, Ratnagiri, Phondaghat
4.	Maval and Ghat Region	Basalt	Vadgaon, Igatpuri
5.	Central dry region	Calcareous	Tuljapur
6.	Wainganga basin of Nagpur Region	Soils derived from mixed Rocks	Amgaon, Shindewahi

Methods of Cultivation

A. Upland cultivation B. Semi dry condition C. Low land (Wet condition)

- | | |
|----------------------|------------------|
| 1. Sowing (Drilling) | 1. Rahu method |
| 2. Dibbling. | 2. Transplanting |
| 3. Furrow method | i) Japanese |
| | ii) Char Sutri |
| | iii) SRI |

A. Up land Cultivation (Dry Condition)

1. Sowing- (Drilling)

Sowing of Rice seed either by seed drill or behind the plough. (Seed rate 100 kg/ha). This method is generally followed in low rainfall region of M.S. This system has low yield potential but required less production cost. Spacing between two lines is 22.5 or 30 cm.

2. **Dibbling-** Seeds are dibbled at 30 x 10 cm (40 to 50 kg/ha), this method is adopted in heavy soils in irrigated region of western region of M.S. The yield and cost of cultivation is high.
3. **Furrow method-** The rice seeds are broadcasted in the furrow opened by Tractor and the sown furrow is compacted by tractor Tire. The method is adopted in heavy, soils of irrigated western M.S. The intercropping on ridges is possible, this method has low cost of production and high net returns.

B) Semi Dry condition- In this method, land is prepared during summer, seeds are broadcasted at the onset of monsoon @ 100 kg/ha Or drilled @ 60 kg/ha Or dibbled @ 20 to 30 kg/ha. During first 45 to 60 days all intercultural operations are completed. Protective irrigation is given whenever necessary.

After regular monsoon, the water is impounded in the field and crop converted in to wet land crop. This method is adopted in areas having moderate rainfall Viz., Orissa, West Bengal, Eastern M.P. and Eastern M.S.

C) Low land Cultivation (Wet Cultivation)

1. Rahu Planting
2. Transplanting.

- i. Japanese method
- ii. *Char Sutri* Method
- iii. SRI Method (*System of Rice Intensification*)

1. Rahu Planting-This system is adopted in Kharlands , where the depth of puddling is more and hence it is difficult to transplant the seedlings. The seeds are sprouted in wet gunny bags , and then sprouted seed are broadcasted in the already puddled field.This method is practiced where there is assured supply of irrigation water.

2. Transplanting- It includes three method viz Japanese method, Char Sutri Method, and SRI (System of Rice Intensification). All these methods required the raising of Seedlings in Nursery.

- Types of Nursery-
1. Wet Nursery
 2. Dry nursery.
 3. Dapog method

1. Wet Nursery-This method of seedling raising is preferred under irrigated condition.Frequently ploughed field is puddled in standing water.Organic manures i.e. about 10-20 tons of FYM or 8-10 tons of green manure is applied and mixed at puddling.The nursery area is brought to fine soft puddle.Area is divided in to small beds of 1.25 m width and with suitable length according to slope. Prepare a drain furrow of 30 cm, width between two bed to facilitates drainage of excess water, sowing, weeding, spraying and irrigation also.Apply 5 kg urea + 10 kg. Ammonium Sulphate + 5 kg. SSP per 100 sq.m. area. The beds are then leveled and about 3-4 cm standing level of water is allowed in bed.Sprouted seeds are used for sowing on these beds. 25 to 30 kg seeds is sufficient for 0.10 ha Nursery and this nursery is sufficient for transplanting of one ha field. Keep the seed bed saturated with water for first five days and then increase gradually the level of water up to 5 cm. Drain out excess water when there is heavy rainfall.

2. Dry Nursery-This method is adopted where availability of water is limited to grow seedlings.Field is brought in to fine tilth by adding 20- 25 tons of FYM/ha and then ploughing – harrowing- clod crushing. Remove the stubbles and make nursery area clean.Open water/drain channel of 50 cm width all around the bed, for watering, intercultural and drainage of excess water. Prepare Raised Beds of 120 cm width and 10- 15 cm. height with convenient length (6 to 7.5 m.). In Chandrapur and Bhandara Districts, due the late onset of monsoon the seedling are raised in fat beds to conserve moisture.The seeds are sown (@ 25-30kg/ 10 R (which is sufficient to transplant 1 ha) at onset of monsoon at 10 cm apart in line in the soil at 2-3 cm depth.2 kg ammonium sulphate and 1 kg urea is

applied for 100 sq.m. nursery area at sowing and same quantity at 15 days after sowing. Seedlings will be ready for transplanting at 21-30 days after sowing. If the field is not ready give same dose of fertilizer to prolong the maturity.

3. Dapog Method of Raising Seedlings.

This method is introduced from Philippines, required less area, save half time in raising seedlings, easy, suitable and most economic. A plastic paper or Banana leaves are used for preparing Dapog Bed. The width of bed is 1.5 m and length as per area available. The plastic paper is spread over wooden board, table, slab or in field also. All four sides of bed are raised with bricks or wooden planks up to 10-12 cm height to keep standing water in bed, take required quantity of seed, dip in water, remove floating seeds which are light and disease affected. Allow the seed to soak in water for 24 hrs by keeping 10 cm standing water above the seed. In this process change the water after every 5-6 hrs. After 24 hrs remove the seeds from water and fill it in wet gunny bag and keep this bag in closed dark room. For better germination pulverize the seeds by hand and openmouth of gunny bag for aeration. After 36-48 hrs the seed will sprout and give out radicals. Spread these sprouted seed on Dapog bed @ 3 kg /sq.m. evenly. In this way about 15 sq.m. area of Dapog bed with about 45 kg seed will be sufficient for transplanting of 1 ha area. Apply water for 3-4 times during a day and keep the sprouted seed wet. Press the sprouted seeds by hand on plastic paper so that the roots will be noted with each other on plastic paper and height of seedlings will be even. This process should be continued for 4 days, after 5th day onward keep the water level in the bed up to 1-2 cm for further 7 days. If the temp are low, cover the bed with gunny bags. Seedlings will be ready for transplanting (11-13 cm height) after 14-15 days from soaking. Roll the bed so that leaves and stem will be inside and roots will be outside. carry the rolled beds to the field, separate the seedlings and transplant.

Land Preparation

For Dry condition- For sowing, dibbling and furrow method of cultivation of rice the basic Cultivation consist of summer ploughing , immediately after harvest of previous crop. The summer ploughing will be help full in following ways. It expose the eggs/pupa of insects, pests and rhizomes of weeds to the bright sunlight. The Exposes provides destruction of all above by bright sunlight or by the birds. Ploughing is followed by clod crushing and harrowing. Before last harrowing add about 10-15 tons/ha. of FYM/ Compost. Spread it uniformly on surface of soil and mix it in soil by last harrowing.

For Transplanting- Plough the land to a 20-25 cm depth in summer. The field will kept flooded or saturated with water for about 15 days before actual transplanting, thus the puddling should be done

15 days before transplanting. This will help in decomposing of weeds, straw and plant parts of previous crop.

Puddling-The processes of making soil impermeable by manipulating and compacting it in standing water, which reduce its apparent specific volume and facilitates transplanting. This is a process by which soil loses its granular structure and becomes deflocculated.

Puddle-A compact mass of earth, soil, clay or mixture of two or more of them, which has been compacted by addition of water and rolling and trampling and made more or less impervious.

Objectives of puddling-

1. It breaks down soil aggregates of clay into their individual particles.
2. It helps to kill the weeds and bury them in puddle soil and reduce the rate of germination of weeds.
3. While Puddling macro pore spaces are destroyed and the upper layer of soil is transferred into fine soft mud or puddle. Thus the soil surface is left in a more even condition which helps for better physical, Biological and Chemical conditions for growth of rice crop.

Puddling operation-The ploughing is done in standing water 2 to 3 times. Soil is stirred up thoroughly by wet land puddler, power tiller or by disc harrow. A green manuring crop is buried in the soil, in standing water. The surface is leveled by harrow beam, collect the stubbles. Add half dose of N and full dose of P and K fertilizers. Now the field is ready for transplanting.

Seed Treatment-3% Brine Solution- 3 kg common salt is dissolved in 100 liters of water, dip the seed in this solution, remove floating seed which is immature, affected by pest and disease. Remove the healthy seed settled at bottom of solution, wash it with cold water and dry it in shed. 1% Mercurial compound- Use Agrosan, Cerasan or Hexasan @2.5 g/kg. to protect the crop from bacterial blight and blast. Azotobacter- Treat the seed with azotobacter culture for enhancing activities of nitrifying bacteria towards atmospheric N fixation.

Spacing and Seed rate

Method	Spacing	Seed rate kg/ha.
Broad casting	-	80-100
Drilling	15, 22.5 or 30 cm.	60-80

Dibbling	Early- 15x15 cm Mid late- 20x15 cm. Late- 20x20 cm	40-50
Transplanting	Early- 15x15 cm Mid late- 20x15 cm. Late- 20x20 cm	20-30
Rahu	-	50-60

Time of sowing- Kharif-June
Rabi- Oct (T.N., A.P. Kerala)
Summer- Feb- March

No of seed/seedlings /hill- 3-4

Depth of sowing / transplanting- 2-4 cm.

Plant population- 3.5 to 4 lakh/ ha

Manures and fertilizers-

Adopt integrated nutrient management. Use Azotobacter as Seed Treatment. Organic manure as FYM / Compost / Green manure at sowing/Transplanting. Use Vermi compost-in Nursery.

Fertilizers-

For early varieties

At planting : Nitrogen- 50 kg/ha, Phosphorus- 50 kg/ha and Potash- 50 kg/ha.

At tillering: Nitrogen- 50 kg./ha.

For mid-late and late varieties-

At planting: Nitrogen- 40 kg/ha, Phosphorus- 50 kg/ha and Potash- 50 kg/ha.

At tillering: Nitrogen- 40 kg./ha

At flowering : 20 kg N/ha

To avoid leaching losses, Nitrogen should be given in form of ammonium sulphate. If the urea is source of Nitrogen, then treat it with neem cake or mix it in compost before applying it in the field. Use Urea or DAP Briquettes for split dose of nitrogen. Use of Bio-Fertilizers viz., Blue-green algae at 10 days after transplanting @ 20 kg / ha Or- 4-5 qt/ha. Of Azola.

Water management :- The water requirement of the Rice is higher than any other crop of the same duration. It requires 800 to 1200 mm of water. The water requirement is high during the initial seedling period covering about 10 Days. Tillering to flowering is the most critical stage to moisture stress. Ensure enough water from panicle initiation stage to flowering (heading).

In transplanted Rice-Water depth of 2 to 5 cm is essential until the seedling established. There after 5 cm of water may be maintained up to the dough stage. Water should be drained out 7 to 15 days before harvesting for quick and uniform maturity of grain.

3. Four Point Technology -It is developed by Dr. N. K. Savant.

1. Control Sowing (15 – 25 X 15 – 25 cm paired rows)
2. Limited use of organic manure. Application of Glyricidia leaves @2.5 tons/ha and application of rice straw @ 1.5 ton/ha while puddling.
3. Application of urea – DAP briquettes in between two rows of rice (15 cm gap)@ 75 kg N + 37 kg P/ha.
4. Use of Rice husk ash @ 1 kg / m² on raised bed of nursery.

SRI Method (System of Rice Intensification)

1. Transplanting of 10-12 days old single seedlings / hill at shallow depth.
2. Spacing 25 X 25 cm.
3. Inter-cultivation / hoeing with the help of kono weeder.
4. Use of organic manures.
5. Proper water management (Alternate wetting and drying).

Weed control :-There is less problem of weeds in transplanted paddy . Weeds should be removed first at 20 days after transplanting. The herbicides found effective in controlling annual weeds.

1. Propanil (stomp f-34) :- spray herbicide @ 3 Kg a i /ha 6 to 8 days after transplanting. Remove all water before spraying of herbicide. The field should be flooded again with in 24 to 48 hrs.
2. Butachlor :- Pre emergence @ 2 kg a i /ha or granules @ 50 Kg/ha in standing water.
3. Fluchoralin (Basalin):- spray solution @ 1 Kg a I/ha or broadcast 50 – 60 Kg /ha granules.
4. Nitrofen (TOK E – 25) :- used as Pre emergence @ 2 kg a i /ha or broadcast 20 – 25 Kg /ha granules within 2 to 3 days after transplanting .
5. Oxydiagryl -(Raft 6%EC, Topstar 80wp) – pre-emergence application for nursery and dry cultivation.

Insects and pests :-Stem borer, Gall midge,Rice hispa,Leaf roller, Army worm, Ear head eating caterpillar,Plant hopper are the important pest of rice.

Stem Borer- Eat inside the stem, causing sterile panicle.

Control measures-

1. Harvesting by Vaibhav sickle.
2. Plough the land immediately after harvesting.
3. Remove all stubbles of previous crop.
4. Avoid late transplanting.
5. Dipping of roots of seedlings in Chloropyriphos.
6. Application of Forate 10G @ 15 kg/ha. 15 days after sowing of seed in nursery.

Leaf Roller- Roll the leaves and eat inside leaves.

Control measure-

1. Change water frequently in the transplanted field.
2. Dip the seedlings in insecticide for 12 hrs before transplanting.
3. Use Pheromone traps for attracting male moths.
4. Spray Malethion when incidence is severe.

Green hoppers and Brown plant hoppers-Eating tender part of leaves. The leaves turned white.

Photosynthesis and finally Yield is affected.

Control measure-

1. Clean cultivation.
2. Transplanting at proper time.
3. Balanced and integrated nutrient management.
4. Spraying of systemic insecticide, when incidence is severe.

Army worms or leaf eating caterpillar.

The insect is nocturnal, always attach on crop during night, cut the plant or panicle at base, creating severe damage.

Control measure-

1. Clean cultivation.
2. Clean the bunds, which is a shelter of this pest.
3. Dusting of 2%Methil parathion dust @20kg/ha. Or spray Cypermethrin during evening.

Diseases :-Blast, Brown spot, Stem rot, False smut, Sheath blight, Bacterial leaf blight, Tungro virus and *Khaira* are the important diseases of rice

Leaf sheath rot/ blight-Spray 50% WP of Bavistin @ 500 gm. In 500 lit. of water first spray at flag leaf initiation and second after 15 days.

False Smut.-Seed and air born disease, causing black panicle.

Control measure-Don't use seed from affected plot. If the climate is favorable for disease, give two sprays of Dithen M-45 orCopper oxychloride at one week interval before panicle initiation.

Harvesting and Threshing :-

The crop should be harvested when about 80 % spikelet's ripened and grains contain about 20 % moisture. The crop cut with serrate edged sickles by hand and left in the field for a few days to dry. Later on harvest is collected in bundles and ready for threshing. Threshing is done by trampling with bullocks or by threshing machine.

Yield :- A well managed crop yields 60 to 70 q/ha. Short duration varieties yields 45 – 55 q/ha

Varieties-

Early-

Name	Duration (Days)	Av. Yield (q/ha.)
Karjat-184	100-105	30-35
Ratnagiri-1	110-115	45-45
Karjat-4	110-115	30-35
Ratnagiri-711	115-120	40-45
Phule Radha	115-120	40-45
Parag	108-112	17-20

Mid-late

Name	Duration(Days)	Av. Yield Q/ha.
Jaya	125-130	45-50
R-P-4-14	120-125	40-45
Phule Maval	125-130	45-50
Palghar-1	125-130	45-50
Phule Samrudhi	125-130	45-50

Late-

Name	Duration (Days)	Av. Yield q/ha.
Ratnagiri-2	140-145	40-45
Karjat-2	140-145	40-45
Mahsuri	140-145	40-45
Suvarna	140-145	40-45
Darana	130-135	35-40

For salty lands(Khar lands)

Name	Duration (Days)	Av. Yield q/ha.
Panvel-1	125-130	35-40
Panvel-2	115-120	35-40
Panvel-3	125-130	35-40

Scented-

Name	Duration (Days)	Av. Yield q/ha.
Basmati-370	130-135	30-35
Pusa Basmati-1	125-130	40-45
Indrayani	135-140	40-45
Bhogavati	135-140	45-50
Prabhavati	110-120	30-35
Avishkar	80-85	30-35
Sugandha	110-115	40-45

Hybrids-

Name	Duration (Days)	Av. Yield q/ha.
Sayadhri	130-135	65-70
Sayadhri-2	125-130	60-65
KRH-2	130-135	70-75
PHB-71	130-135	65-70

2. Pearl millet (BAJRA)B.N. *Pennisetum glaucum* L.Family :- Poaceae

Production in world: It covers 60% area is in Africa., 35% area is in Asia, 4% in European countries and 1% in north America.

Area and Production in Asia

Country	Area in '000' ha	Production in '000' tons
China	1100	2200
India	9800	9400
Myanmar	250	150
Nepal	258	253
Pakistan	343	193

Area and Production in India

State	Area (million ha.)	Production (million tons)	Productivity q/ha
Rajasthan	3.2	42.1	7.88
Maharashtra	1.55	20.4	6.73
Gujarat	0.94	12.4	12.77
U.P	0.7	9.2	12.35
Haryana	0.52	6.8	13.31
Karnataka	0.31	4.1	5.32
M.P	0.15	1.9	11.50
Tamilnadu	0.12	1.6	11.44
A.P	0.09	1.2	8.72

Area and Production

Particulars	India	Maharashtra
Area (M. ha.)	10.45	1.93
Production (M. tones)	6.91	1.14
Av. Grain yield (q/ha.)	6.73	15.78

Economic Importance

1. Next to sorghum, pearl millet is important staple food crop of India.
2. It is considered as a poor mans food grain.
3. Among all the millet crops, pearl millet stands first position and it ranks fifth in order of cereal crops.
4. It is nourishing food and its nutritive value is considered to be comparable to the rice and wheat.
Bajra grain contains
 - i. Carbohydrates – 67.1 %, ii. Protein – 11.6 %, iii. Fat – 5.0 %, iv Fiber – 1.2%,
 - v. Minerals- 2.7%, vi Moisture –12.4%

Origin and History :-The home of the pearl millet is said to be Africa from where it spread to India. The cultivated race are reported only from the Tropical Africa.

Botanical Description :-

Root system :-The root system of bajra is fibrous, at the time of germination a primary root develops from seed and penetrate deep in the soil. Primary root consist of several thin and fine lateral roots. When seedling reaches 2 to 3 leaf stage, the secondary root develop from the bud located on lower most node near soil surface.

Stem- The stem is solid and single, but often they are branches , both primary and secondary. It is made up of nodes and internodes.

Leaves :- leaves of bajra are long and fairly broad and held erect by thick midrib. Leaves arise alternately on stem. They are smooth on the lower surface and rough on the upper surface. leaves consist of a blade and sheath.

Inflorescence-The inflorescence of bajra is almost a cylindrical head densely packed with the spike lets. The number of spike lets may vary from 800 to 3000 per head. the length of head may vary from 15 to 40 cm. It is a cross pollinated crop.

Caryopsis (Grain)- The seed is caryopsis, and is almost oval with one of its ends tapering . The seed is 3-4 mm long and 2-2.5 mm wide. The color of the grain varies from whitish – yellow to gray or dull light blue

Climate

Bajra is a tropical crop, requiring warm and dry climate for its proper growth. It can withstand drought to a greater extent and hence it is suitable for area receiving 500mm rainfall. During the vegetative growth moist weather and light to medium rainfall is very useful, while at flowering and grain formation there should be no rain and weather should be clear and dry. The average temperature requirement during the growing period is 26°C to 32 °C

Soil :-Bajra can be grown on variety of soils from sandy to medium black soil and alluvial soils. Sandy loam to loamy soil are most suited, but these soils should be well drained. Bajra is sensitive to water logging condition and acidic soil. In Maharashtra it is also grown in light type of soil.

Preparation of land –

Bajra crop needs very fine and compact seedbed, because the seed size of this crop is relatively very small hence one ploughing (up to 10 – 15 cm deep) followed by 2 to 3 harrowing will be sufficient. Add 5 tones of FYM or Compost before last harrowing.

Seed and Sowing-

Time of sowing-

The best time of sowing for bajra is immediately after receiving good rains and up to 15th July, delay in sowing reduces the yield to the considerable extent, due to heavy incidence of downy mildew and ergot diseases.

Seed Rate- Hybrid varieties – 2.5 to 3 Kg /ha.
For local improved varieties - 5 Kg /ha.

Method of sowing-

Sowing of bajra is carried out by drilling method with three coultered seed drill, The bajra seeds are very small in size , it should be mixed with equal quantity of sand which facilitate uniform sowing of the seed. Also sown behind plough to conserve moisture.

- Depth of sowing- not more than 2 to 3cm
- Spacing- 45 x 10 cm or 45 x 15 cm
- Plant population- 1.60 to 1.80 lakh / ha

Seed Treatment-

The seed should be treated with 20 % brine solution for avoiding the incidence of ergot disease of pearl millet. To control Downy mildew disease of pearl millet seed should be treated with either Thiram/ Captan @3 g/kg or with Apron @100 g / kg seed. Finally treat the seeds with Azospirillum culture @ 250 g / 10 Kg seed.

Interculture :-

Thinning – The first thinning at 10 DAS and second thinning at 20 DAS. Tillering starts from 20 to 25 days after sowing and continuous for 15 days. Keep only one plant at about 10 to 15 cm.

Hoeing- Two hoeing at an interval of 15 days up to 45 days

Weeding- Two weeding, the crop should be kept weed free up to first 30 days.

Chemical weed control- pre-emergence spray of Atrazine @ 0.5 Kg a i /ha is effective to control weeds up to 30 to 35 days

Fertilizers-

For light type of soil -

40 Kg N + 20 Kg P₂O₅ + 20 Kg K₂O kg /ha.

For medium to heavy type of soil –

50 Kg N + 25 Kg P₂O₅ + 25 Kg K₂O kg./ha.

Half of the nitrogen and full dose of P and K was applied at the time of sowing remaining N will be given at 21 days after sowing

Irrigation Management-The total water requirement of the bajra crop is 25 to 30 ha cm including rainfall. Bajra is mainly grown as rainfed crop. Only 3 to 7 % of the total area under this crop is under irrigation. If there is long dry spell, during any critical stage of crop growth, then the one or two protective irrigations are given.

The critical growth stages of bajra are-

Maximum Tillering stage 25 – 30 DAS

Flag leaf and flowering stage 40 – 45 DAS

Grain filling stage 60 – 65 DAS

Cropping systems-

1. Bajra – Fallow
2. Bajra –Groundnut,
3. Bajra –Groundnut – jowar,
4. Bajra – Ragi .

Insect and Pest-

1. Shoot fly-The late sown crop is damaged by shoot fly. In seedling stage, the shoot dies and plant growth is restricted.

Control measures- Early sowing, Seed treatment, two sprays of any contact insecticide at 7 and 15 days after emergence.

2. White Grub- The adult larva living under ground, damage the crop by eating roots. The patches of dry plants are observed in the field.

Control Measure- Use only well decomposed FYM. Deep ploughing in summer to expose insect to sunlight and Birds. Destroy the Adult Grubs by collecting them in early night from their shelters i.e. Babul and Neem trees. Apply contact insecticide in powder form in the soil at early stage of larva (First week of June).

3. Gray Weevil-Feed on tender plant parts.

Control Measure- Clean Cultivation. Use of contact as well as systemic insecticide at early stage of insect.

4. Blister Beetle- Adult insect feeds on grains at milk stage. It releases acidic liquid from body for self protection. Which may be harmful for human being.

Control Measure-

- Spray neem extract on ear heads, which will act as a Repellent.

- Use any dust of contact insecticide

Diseases-

1. Downy mildew-Yellowing of plant. Stunted plant growth. Panicle emerged as grassy head.

Control Measure-Seed treatment.

2. Ergot- The complete head damaged with considerable loss in yield.

Control measure-Brine solution seed treatment.

3. Smut- This is seed and air born disease. Instead of normal head a black powdery panicle emerged from shoot.

Control measure- Seed treatment with sulphur, Destroy the infected plants, avoid using seed from infected field.

4. Rust- Red and brown powdery spots appears on lower side. Reduce the photosynthesis and finally yield. **Control measure**-Early sowing, drain out excess water from field. Spray Mancozeb @ 1.5 kg / ha in 500 lit. water.

Harvesting- It require in 85 to 90 days to mature

Sign of maturity :-

- 1) The grain reach the hard dough stage or moisture % of the grain 15 - 20 %.
- 2) Grains give cracking sound when crushed under teeth.
- 3) Yellowing and drying of the older leaves

Harvesting is done by cutting the entire plant or removing the ear head first and cutting down plants later. The ear head after harvesting should be sun dried at the threshing floor for 10 to 20 days. The stalk are tied in small bundles.

Threshing :- Ear heads are trampled under the feet of bullock or threshing was done by power thresher cum winnower.

Grain Yield :- local and improved varieties 20 – 25 q / ha

Hybrids 30 – 35 q / ha

Dry fodder yield- 35 to 50 q / ha

Improved Varieties - ICTP 8203, WCC 75, ICMV – 221, ICMV – 155, Dhanshakti

Hybrids :- Shraddha (RHRBH 8609), Saburi (RHRBH 8924), Shanti (RHRBH 9808),

Adishakti (DHBH-9071)

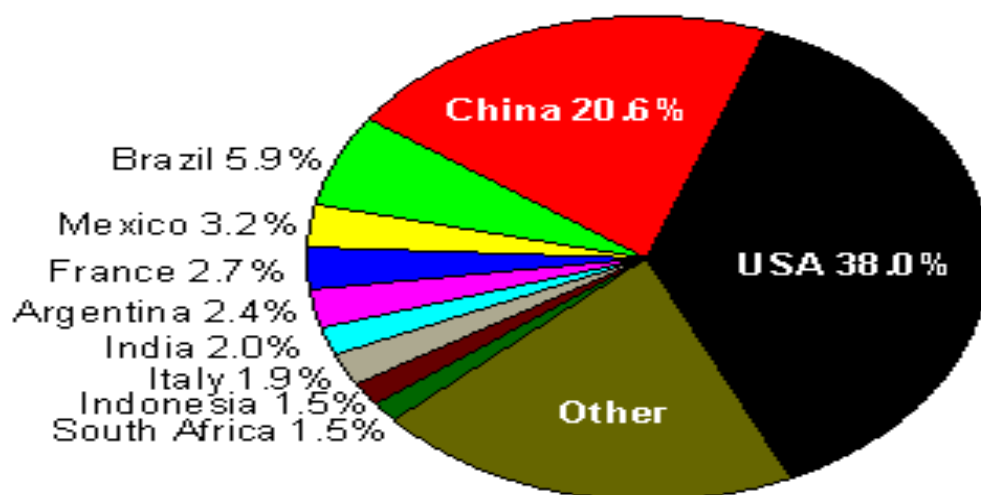
3. Maize (Maka) B.N. *Zea mays* L. Family :-Gramineae / Poaceae

Origin and History :-Maize is a native of South America (Mexico) and introduced to India by the Portuguese during the 17th centuries.

Area and Production in World-

Area about 139 m ha with total production about 600 million tones. About 40 % production is from US followed by 21% from China.Total 58 % production in developed countries and 22% in developing countries.Wide adaptability- every month the crop is harvested some where in world.It grows from 58⁰ North latitude (in Canada and Russia) to 40⁰ south latitude (Southern Hemisphere)Below MSL to at altitude of more than 3658 m. In region receiving less than 250 mm annual rainfall (semi arid plains of Russia) and also in regions having more than 1000 mm rainfall in Pacific cost of Colombia.

Maize production



Area and Production in India-

A 6 million ha. Area as a rainfed crop, only 22.8% is under irrigationGrown through out the country.

Area and production of major maize growing states is as given below

State	Area(000 ha)	Prod. (000 ton)	Av. Yield (q/ha)
Rajasthan	946.4	1024.7	10.83
Uttar Pradesh	901.3	924.7	10.26

Madhya Pradesh	849.8	1180.5	13.89
Bihar	685.6	1270.3	18.53
Karnataka	498.0	1618.0	32.45
Gujarat	408.3	696.0	17.05
J & Kashmir	311.4	532.4	17.10
Himachal Pradesh	305.2	671.3	22.01
Maharashtra	278.4	511.4	18.37

Classification :-

Maize is divided in seven groups. The classification is based on character of the kernels

1. *Zea mays indurata* or flint corn – Outer portion of kernel is composed of hard starch which restrict absorption of moisture and avoid spoilage. It mature earlier.
2. *Zea mays indentata* or Dent corn- Mostly grown in U.S. The hard starch confined to the sides of kernel. The soft starch which form the core and cap, contrast when dried and produce dent in the top of kernel. Used to feed live-stocks.
3. *Zea mays everta* or Pop corn- Popular snack food. Kernels are small. When heated at 170⁰ C the grain swell and burst turning inside out.
4. *Zea mays saccharata* or Sweet corn- Used as Raw boiled or roasted, harvested with 70% moisture, mainly grown in U.S. and Canada. It is good source of energy comprising 20 % sugar and Vitamin C and A.
5. *Zea mays amylacea* or Soft corn- The kernel is composed with soft starch and hence easy to grind and make flour. Mostly grown in drier area of south west U.S.
6. *Zea mays tunicata* or Pod corn- The size of cob is very small.
7. *Zea mays Ceratina Kulesh* or Waxy corn- Waxy appearance of kernel. China is a origin of this gene.

Economic Importance-

- Grain - As a food (Contain Protein – 10%, Oil – 4 %, Carbohydrate – 70%, Crude fiber – 2.3%, Albuminoides – 10.4%, vitamin A & E, Nicotinic acid and Riboflavin in view of above characteristic maize is called as Queen of cereals
- The special variety called pop corn the grain of which are converted into popped.
- As a cattle, Poultry and piggery feed.
- In industries as a raw material for preparation of starch syrup, alcohol, acetic acid, lactic acid, glucose, plastics, rubber, ethanol, etc.
- Maize oil is very fine cooking medium when fully refined.
- **Cob** -Green cobs are roasted and eaten by people with great interest.Special varieties of Baby corn grown for Baby corn which is used in various Indian as well as continental dishes.
- Fodder-Green fodder is used for feeding milking animals

- It can be grown in all the year round i. e. *Kharif, Rabi, and Summer* season.

Botanical Description-

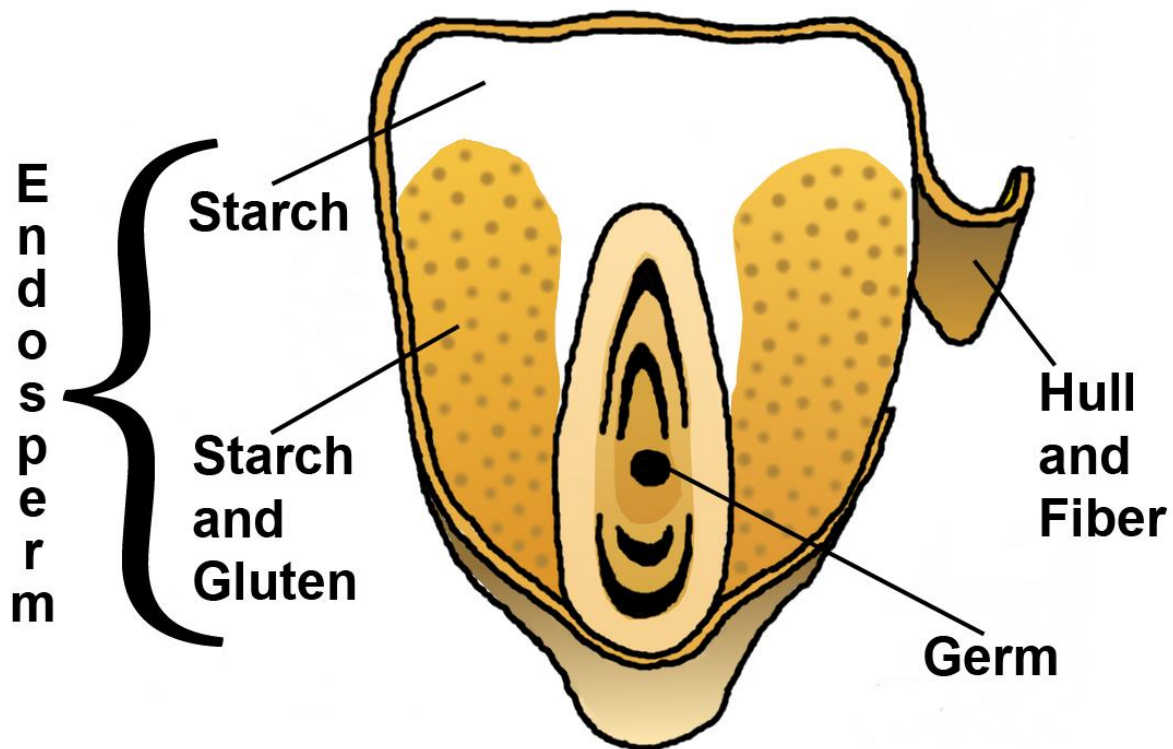
Roots- Primary or tap root- develop in form of radical. Seminal roots- present in un-germinated seed and functional during initial phase.(Temp. roots).Adventitious roots- permanent roots, Shot born roots, Develop on basal node of stem below the soil.Prop or Brace roots- Shot born roots, Develop on lower most 2-3 node of stem above the soil. (support to plant)

Leaf-The leaf portion attached with stem is leaf sheath and the other is called leaf blade. The boundary between sheath and blade is a ligules

Tassel- Branched inflorescence, consist of central spike and 10-50 lateral branches. The paired spike lets occur around the central spike. Each spike let contain 2 florets. The upper florets developed first. Each floret is enclosed with lemma and palea. Each floret contain 2-3 anthers At anthesis the lodicules joining the lemma and palea swell and allow extruction of anthers by elongation of filament.

Fertilization-Female flower contain single ovary terminated by long style known as silk. The silks are covered with fine hairs used to capture wind blown pollens. Pollens are produced at tassel (30-60 million/plant). After pollination the pollen tube reach at the ovary with in 15-25 hrs. The end of tube ruptures and release two sperms, one of these fuses with female egg nucleus to produce embryo and other fuse with a double nucleus to produce endosperm a food organ.

Ear-Born at auxiliary buds, capable of producing mature kernels. Each auxiliary bud is covered with 8-10 modified leaves called as husks and a prophyll. The thick axis of ear is called cob, similar to central spike of tassel with multiple rows of paired spike lets.



Climate :-Maize is mainly grown in tropical and temperate region. However it is essentially a warm and humid season crop and in areas of mild climate it can be grown through out the year. It can be successfully grown where the night temperature does not go below 15.6°C . The most suitable temperature for germination is 21°C and for growth is 32°C . It cannot withstand frost at any stage of its growth. Maize can be successfully cultivated in the areas receiving rainfall more than 750 mm.

Soil :-Maize can be grown on wide ranges of soils. However, maize grows best on fertile, well drained medium to heavy textured soils. It can be successfully grown in the soils where pH ranges from 5.5 to 7.5. The crop grown on river banks (alluvial soil type) especially gives good yield of maize.

Preparatory tillage :-Ploughing (up to 20 – 25 cm deep) followed by 2 to 3 harrowing. Before last harrowing add 12 to 15 tones of FYM or Compost. If maize is to be grown under irrigation for grain purpose the ridges and furrow are opened at the distance of 60cm and 75 cm apart in medium and heavy type of soil, respectively.

Methods of sowing :-

1. Drilling
2. Dibbling on ridges and furrow

3. Putting the seed in plough furrow

Spacing :-

Mid late to late varieties- 75 X 20 to 25 cm

Early and very early varieties - 60 X 22.5 cm

Fodder – 30 X 10 cm

Depth of sowing :- Not more than 5 cm

Seed Rate :- Grain - 15-20 Kg /ha

Fodder - 75 Kg / ha

Manures and fertilizers-Maize is heavy feeder crop.Add 12-15 tones of FYM/Compost at last harrowing.

- Apply NPK @ 90:40:40 for rain fed crop.
- Apply NPK @ 120:60:40 for irrigated crop.
- Apply NPK @ 120:40:20 for fodder crop. (80kg N+ full dose of P and K at sowing and 40kg. N at 30DAS)
- For grain crop apply 1/3rd dose of N and full dose of P and K should be applied at sowing.
- 1/3rd dose of N at 30 DAS.
- 1/3rd dose of N at 50 DAS.
- Zink Sulphate @ 25kg/ha (If deficiency observed)

After Care-Gap filling within 8 to 10 days after germination.Thinning is carried out 10-15 days after sowing. During thinning only one healthy plant is kept at each hill.Weeding and hoeing- Growth at initial stage is slow and hence 2-3 hand weeding and 2 hoeing should be given at an interval of 12-15 days.Chemical weed control :- Effective weed control can be obtained by pre-emergence spray of Simazine / Atrazin @ 1 Kg a.i./ha.

Irrigation Management-

The water requirement of maize on account its broad leaves and tall growth is very high. The total water requirement of this crop is 60 ha cm. Maize does not tolerate water logging. The early vegetative stage(20-40 DAS), Tasseling and silking stages (45-60 DAS) are critical growth stages for water

Insect and Pests

- **Stem borer, Pink stem borer, Shoot fly**- Damage the crop at early stage- Spray any contact insecticide at seedling .
- **Maize leaf roller, Kharif grass hopper**- feed on the foliage- dusting of contact insecticide.
- **Cut worm, White grub** – damage the crop at early stage- soil application of insecticide.

Diseases:-

- Seed rot and Seedling blight- Seed treatment
- Bacterial stalk rot and Pythium stalk rot- Seed treatment
- Sugarcane downy mildew, Brown strip Downy mildew- Spray M45 @ 500 ml in 250 lit. of water for two times .

Birds- (Parrots)

- Treat the cobs of border rows with 5% Malathion dust.
- Avoid planting near trees, electric wires and buildings.
- Adopt Cluster planting rather than isolated one.

Cropping systems and rotation-Maize crop must be rotated with legumes in order to maintain soil fertility. Green gram and Black gram are the ideal inter crops grown with maize in the ratio of either 2:1 or 4:1

Varieties-

- **Improved-** Selection made from local (Indigenous) cultivars.
- **Composite-** Produced by mixing seed of several outstanding lines and encouraging open pollination to produce crosses in all combinations among themselves.
- **Hybrids-** These are first generation (F1) from crosses between two pure lines. It may be single hybrid or double hybrid.

Late Maturing :-		
Hybrids :-	Duration(days)	Yield(q/ha)
Ganga-11	100-110	50-60
Deccan-105	100-110	50-60
Trishulata	100-110	55-60
JK-2492	100-110	55-60
Pro-311	100-110	50-60
Rajarshi	100-110	50 -60
Composite varieties		
Dhaval	100-110	45-50
African Tall	120-125	40-50

Mid late		
Hybrids-		
DHM-107	90-100	50-55
KH-9411	90-100	50-55
MMH-69	90-100	50-55
Composite		
Manjari	90-100	40-50
Navjot	90-100	40-50
Prabhat	90-100	40-50
Karveer (MPQ-13)	95-100	52-55

Early -

Hybrid	Duration	Av. Yield (q/ha)
Deccan Hybrid-1	80-85	45-50
Deccan Hybrid-109	80-85	45-50
Pusa Hybrid-1	80-85	45-50
Pusa Hybrid-2	80-85	45-50
MMH-133	80-85	45-50
X1123G (3342)	80-85	45-50
Composite		
Kiran	80-85	45-50
Panchganga	80-85	45-48
Pop Corn Varieties-		
Amber Pop Corn	90-100	20-25
Sweet Corn Varieties		
Madhuri	70-80	40-50 Gr. cobs

Harvesting :-The maize crop raised for grain is harvested when the grains are nearly dry do not contain more than 20% moisture. Sheath turns brownish and grain become hard. Ear or cobs are removed from the standing crop. Harvested ears/ cobs are dried for 2 to 3 days in the sun before shelling.

4. Sorghum (Jowar) Great millet, B.N :- *Sorghum bicolor* L. monech

Economic importance:-

- It is commonly known as great millet due to large size of grain among millets and large area under cultivation.
- It is fourth most important world cereals following wheat, rice and maize it grows well in variety of soils.
- It is staple food crop of drier parts of Africa, India and China.
- The grain flour is used for preparation of
- Ear head of some sweet varieties are eaten as *Hurda*.
- The grain of some varieties is popped and consumed as pop grain.
- Jowar grains are also used for manufacturing starch, syrup, glucose, sugar tablet, oil, gluten and as a poultry feed.
- Grain contains 70 % Carbohydrates, 3 % Fat and 9 to 11 % protein.
- The dry stalk and leaves called *kadba* are used as fodder for cattle or also grown for green fodder purpose.
- Jowar or sweet sorghum stalks are used for preparation of Jaggery.

Area and Production in World.

Country	Area (000 ha.)	Production (000 ton)	Productivity (qt/ha)
Asia	18.451	18.867	10.23
Africa	17.799	12.784	7.18
North and Central America	5.970	21.325	35.72
South America	1.353	3.537	26.14
Oceania	0.407	0.934	22.98
World (1990)	44.352	58.190	13.12

Botanical description- Genus- Sorghum, Family- Poaceae

A- Head - The inflorescence is a loose to dense panicle, having many primary branches borne on a hairy axis bearing paired spikelets.

B- Leaves- arises from each node and is about 60 to 100 cm long and 2.5 to 8.5 cm broad.

C- Stem-Groove on the inter node, groove arranged alternately from one side to another on each successive inter node.

D- Flower- There are two flowers in fertile spikelets, the lower is sterile and upper is fertile. Each panicle contains on an average 45g seed. Sorghum is a often cross pollinated crop.

E-Roots- Roots are superficial and fibrous , Adventitious roots emerge from immediate upper node and provide support to the plant.

F- Grain- The seed colour is varies from white to dull white.

Each panicle contains on an average 45g seed. Sorghum is a short day plant

Origin and History-

It is a native of eastern Central Africa, from where it is spread to India hence it is considered that Africa and India may be the original homes of the sorghum.

Climate :- It is purely a tropical crop, during Kharif, it is cultivated in areas where the annual rainfall is 600 to 1000 mm usually distributed between last week of June to the first week of Sept. The average temp. during the growing season should be 26°C to 32°C. It can tolerate with minimum Temp 16°C and maximum Temp 40°C. It is a drought resistant crop. It regulates transpiration and remains dormant during the drought period and starts active growth when the moisture status of the soil improves. Excessive soil moisture as well as prolonged drought are harmful for crop growth.

Soil :- Medium to deep black soils are predominantly suitable for growing sorghum. In areas where there is assured rainfall it can be cultivated on light soil. It tolerates saline and alkaline conditions of the soil up to some extent (pH 6.0 to 8.5)

Preparatory tillage - Sorghum requires firm and compact seedbed. Immediately after harvest of the previous crops, the land should be ploughed up to the depth of 15 – 20 cm. However, ploughing is not necessary every year, if previous crops are not infested with perennial weeds, the land is prepared by giving 2-3 harrowings which are sufficient. Before last harrowing add 10-15 tones of FYM / Compost in the field and mix with the soil.

Seed treatment –

- The seed is also treated with 30 % brine solution for avoiding the incidence of ergot disease of jowar
- The seed is treated with 300 fine mesh sulphur @ 4 g / Kg seed for the control of loose smut disease of jowar.
- The seed is treated with Carbofuron @ 10 g / Kg seed for the control of shoot fly attack during early crop growth stage
- The seed is also treated with Azotobacter culture @ 250 g / 10 Kg seed

Sowing-

Sowing Time-Immediately after commencement of monsoon and after having adequate moisture in the soil before 7th July. Delay in sowing reduces soil temp. and it will affect the germination. Delay in sowing also increases the incidence of shoot fly.

Method of sowing :- Drilling and dibbling method

- Spacing - 45 X 15 cm
- Depth of sowing - Not deeper than 5 cm

Seed rate –

Hybrid and high yielding varieties :- 8 to 10 Kg /ha

Plant population :- Hybrid varieties 1.80 to 2.25 Lakh / ha

Manure's and fertilizers

Apply FYM or compost @ 6 to 8 tones/ha. for Rai fed crop and 12 to 15 tones/ah. for Hybrids at last harrowing.

Fertilizers

- For light soil :- 50 Kg N + 25 Kg P₂O₅
- Medium soil :- 80 Kg N + 40 Kg P₂O₅ + 40 Kg K₂O
- Heavy soil :- 120 Kg N + 60 Kg P₂O₅ + 60 Kg K₂O

Half of the nitrogen and full dose of P and K will be given at the time of sowing, Remaining half nitrogen after 1 month of sowing

Inter culture operations-

- Thinning- first at 10 to 12 DAS.second at 20DAS.
- Weeding and hoeing- Within 30 DAS,2 hoeing at 10 Days interval and one weeding is given to control the weed
- Chemical weed control-1. Pre emergence spray of Atrazine @ 1 Kg a.i./ha 2. Post emergence spray of 2 , 4 -D @ 3 to 4 Kg a i./ha

For striga- 2 , 4 -D @500 gm + 0.5% paraquat 500 ml. in 500 liter of water should be followed with 2 to 3 spray at 15 days interval. Or fallow the proper crop rotation

Irrigation management

The total water requirement of *Kharifjowar* is 40 to 50 ha cm depending upon the variety and soil type.

The critical growth stages of Sorghum are-

1. Initiation of grand growth stage : 20 to 25 DAS
2. Flag leaf stage or Boot stage : 50 to 55 DAS
3. Flowering stage : 70 to 75 DAS
4. Grain filling stage : 90 to 100 DAS

Sorghum Based Cropping Systems-

- Crop rotation-
- Sorghum – pulses,
- Sorghum – chilies,
- Cotton – Sorghum.
- Intercropping- intercropping with Green gram, groundnut, pigeon pea in 4:2 or 2:4.

Sequence cropping- Gram, Safflower

Plant Protection-

Insect pests-

1. **Shoot fly, Stem borer**-In seedling stage, the shoot dies and plant growth is restricted.

Control measures- early sowing, Seed treatment, two sprays of any contact insecticide at 7 and 15 days after emergence.

2. **Army worms and ear head caterpillar**- Both the pests make damage by eating plant parts and immature grains.

Control measures- Dusting of any contact insecticide in powder form.

Diseases –

1. Grain smut- This is seed and air born disease. Instead of normal head a black powdery panicle emerged from shoot. **Control measure-** Seed treatment, Destroy the infected plants, avoid using seed from infected field.

2. Grain Mould- Heavy or continuous rains at maturity with high humidity is favorable for development of grain mould. the quality is destroyed. **Control measure-** Seed treatment, destroy the infected plants, avoid using seed from infected field. Three sprays of fungicide after flowering.

Harvesting- It require 105 to 135 days to mature.

Sign of maturity-

- 1) The lower grains of ear head become hard and when crushed under teeth give cracking sound & breaks into pieces
- 2) The grain moisture content at this stage varies from 25 – 35 %
- 3) A blackish spot develops at the tip of the grain
- 4) yellowing of the peduncle
- 5) yellowing and drying of the older leaves

Yield :-	Grain	Straw
Kharif Rain fed	8-10	12-15
Kharif high yielding	25-30	40-50
Hybrid (irrigated)	40-50	60-70

Improved Varieties

Name	Duration (Days)	Grain yield (q/ha)	Fodder yield (q/ha)
SPV-462(CSV-12)	110-115	35-40	60-75
SPV -475 (CSV-13)	110-120-	35-40	60-75
SPV- 946 (CSV-15)	110-120	40-45	80-100
CSV-17	110-120-	35-40	60-70
PVK-801	110-120	35-40	60-70
CSV-23	110-120	35-40	60-75
Phule Amruta (RSSV-1)	110-115	40-45	80-100

Hybrids-

Name	Duration (Days)	Grain yield (q/ha)	Fodder yield (q/ha)
CSH-1	100-105	30-35	40-50
CSH-5	110-115	40-45	100-110

CSH-6	95-100	40-45	80-90
CSH-9	110-115	40-50	110-120
CSH-13	110-115	35-45	100-115
CSH-14	110-115	35-45	110-115
CSH-16	110-115	35-45	100-115
CSH-17	110-115	35-45	110-115
CSH-18	110-115	35-45	115-120
CSH-21	110-115	35-45	110-115
CSH-23	110-115	40-45	110-115

Sweet Sorghum-SSV-84, HES-4, CSV-19SS, CSH-22SS, AKSSV-22

Jaggery variety - SSV-84

Local Varieties :-Saonar, Ramkel, NJ-164, NJ-15 (Vidharba region)

Aispuri, Satpani, Khadi, (Jalgaon & Dhule)

Dukari for fodder (Satara, Sangali and Kolhapur)

5. Small or Minor millets-

Sr. No.	English name	Common name	Botanical name
1.	Finger millet	Ragi (<i>Nachani</i>)	<i>Eleusine Coracana L</i>
2.	Foxtail millet	Kakun (<i>Rala</i>)	<i>Setaria italia L.</i>
3.	Kodo millet	Kodra	<i>Paspalum scrobiculatum L.</i>
4.	Little millet	Kutki (<i>Sava</i>)	<i>Panicum sumatrense</i>
5.	Proso or Common millet	Cheena (<i>Varai</i>)	<i>Panicum miliaceum L.</i>
6.	Barnyard millet	<i>Bhagar</i>	<i>Echinochola frumentacea</i>

Nutritive values of different millets

	Protein (g)	Carbohydrates (g)	Fat (g)	Minerals (g)	Fiber (g)	Calcium (mg)	Phosphorous (mg)	Iron (mg)	Energy (Kcal)	Thiamin (mg)	Niacin (mg)
Finger	7.3	72	1.3	2.7	3.6	344	283	3.9	336	0.42	1.1
Sorghum	10.4	70.7	3.1	1.2	2.0	25	222	5.4	329	0.38	4.3
Pearl	11.8	67.0	4.8	2.2	2.3	42	-	11.0	363	0.38	2.8
Foxtail	12.3	60.2	4.3	4.0	6.7	31	290	2.8	351	0.59	3.2
Little	7.7	67.0	4.7	1.7	7.6	17	220	9.3	329	0.3	3.2
Kodo	8.3	65.9	1.4	2.6	5.2	35	188	1.7	353	0.15	2.0
Proso	12.5	70.4	1.1	1.9	5.2	8	206	2.9	354	0.41	4.5
Barnyard	6.2	65.5	4.8	3.7	13.6	22	280	18.6	300	0.33	4.2
Paddy Rice	6.8	78.2	0.5	0.6	1.0	33	160	1.8	362	0.41	4.3
Wheat	11.8	71.2	1.5	1.5	2.0	30	306	3.5	348	0.41	5.1

Origin-The finger millet is probably native from India and from here spread to Africa. Other small millets are mostly originated in Asian or African continent.

Area and production in India-Area under small millet declined from 7.7 million ha. In 1949-50 to 4.0 million ha. In 1994-95. However, the production remain same as in 1950 i.e. 3.7 million tone. Finger millet contribute about 50% area and 80% of total production of small millets.

Economic Importance-

- Capability of growing under adverse soil and climatic conditions.
- In mid season correction (late rains, failure of first crop or flood conditions.)it gives good yields rather than no crop.
- First crop that is harvested in *kharif* season provide food grains in lean months.
- Like other cereals the grains are source of protein, carbohydrates and minerals.
- Staple food of tribal population living in hilly regions.

Botanical Description-

1.Finger millet- (Ragi / Nachani)

It is an erect annual plant grows up to 135 cm. height. Tillers branches profusely. Stem is compressed and elliptic. Leaves are linear with distinct midrib. The leaf sheath envelops the stem. The panicles are in 4 shapes curved top, incurved, open and fish shaped. The length of finger vary. Each finger contain about 67-73 spike lets. Each spikelet contain 4-6 flowers. The opening of florets is from bottom to top. Time of opening is from 1 AM to 5AM. It is a self pollinated crop. After fertilization each spikelet contain about 3-8 small and brown colored seeds.

2. Foxtail millet- Kakun (*Rala*)

Grows up to 60-75 cm height. Leafy stem, bends at maturity, leaves are narrow and green in colour. The inflorescence is dense, cylindrical and bristly. Spikelet are 2 flowered generally in cluster of 40 to 50.

3. Kodo millet – (Kodra)

It is erect annual plant with hairy nodes grow up to 45-90 cm. height. Profusely tillered, The leaves and stem are purple in colour. The inflorescence is a panicle with 2-8 spikes. Spikelet has a single self pollinated flower open in early morning

4. Little millet- (Kutki / Gunduli / *Halavi varai*)

It has erect and slender stem with swollen nodes grows up to 45-100 cm height. Leaves are linear and slender. The inflorescence is large panicle with numerous spikelets with hermaphrodite flowers. The seed is caryopsis and may be creamy white, yellow or red in colour.

5. Porso millet / Common millet- (*Cheena / Varai*)

It is a erect grows up to height of 75-100 cm. purple pigmentation may be observed on plant parts. Stem is slender and leafy up to panicle. The inflorescence is 14-40 cm long oblong panicle having erect hairy branches. Grains are olive brown in colour.

6. Barnyard millet- (*Sava*)

It is tall erect plant grows up to 50-95 cm height. Stem and leaves are green in colour. Inflorescence is narrow, pyramidal, crowded with unawned spike lets in 3 rows. The flowers are hermaphrodite. The grain is caryopsis and white or yellow in colour.

Soil –It can be grown on all types of soils. Tolerate slightly alkaline soils. Best soils are of alluvial, loamy and sandy types having better drainage. Heavy, black cotton soils with poor drainage and stony soils with poor fertility are not suitable.

Climate –It is tropical and sub- tropical and heat loving crop, the best temp. range is 26-29°C with annual rainfall of 500 to 900 mm. In high rainfall area it is grown as transplanted crop.

Preparation of land –Ploughing and harrowing for dry cultivation and transplanted crop of Ragi. Light ploughing with planking is sufficient for other small millets. Generally small millets are

grown on marginal lands on stored moisture, deep ploughing in summer will be helpful to conserve rain water.

Time of Sowing- June – July (Immediately after onset of monsoon)

Spacing- 20-25 cm line spacing for Ragi
Broadcasting for other small millets.

Seed rate- Ragi- 6-8 kg / ha. Other small millets- 8-15 kg / ha.

Manures and Fertilizers- Ragi- 60:30:30 kg NPK / ha.

Other Small millets- 20-40 kg N + 10 - 20 kg P₂O₅ + 10-20 kg K₂O / ha

Weed Control- Keep crop weed free for first 45 days either by hand weeding or with the use of chemicals. (Metoxuron @ 0.75 kg. a.i./ha)

All minor millets are generally grown as mixed crop with sorghum, Bajra, pulses and oil seeds in *kharif* season under rainfed conditions. When grown on marginal land in hilly regions, sole crop is taken in *kharif* season.

Plant Protection-

Insect pest are- stem borer, hairy caterpillar, hoppers, and ear head eating caterpillar.

Diseases- Ragi- Blast, Blight, downy mildew.
Foxtail millet- Smut
Kodo millet- Ergot
Porro millet- Head smut
Barnyard millet- Downy mildew

Varieties-

Finger millet - **Early- (90-102 Days)- Av. Yield is 2-3 q / ha.**

PES-400, VL- 149, KM-65, GPU- 26, VR-708,

Mid late- (110-120 days)- Av. Yield- 3-5 q / ha.

Dapoli-1, Phule Nachani, GPU-28, Co-13, Indaf-5, A-404,
Gautami, RAU-8, HR-374.

Late- Above 120 days- Av. yield- 3-5 q/ha.

PR-202, GN-3, Indaf-8, PES-110,

Proso millet : Early : K-1, Mid late : GPUP – 21, Phule Ekadashi

Barnyard millet : Mid late : VL-29

Foxtail millet : Arjuna, PS-4

Little millet : KOPLM-83, Dindori-1

Kodo millet : Nivas-1, RBK-155, GPUK-3

In case of all the millets local types are also used.

Harvesting and Threshing-All minor millets mature within 100 to 135 days. At maturity the straw and head turned to yellow, the grains become hard. Cut the plants at ground level, dry it in field for 3-4 days. Threshing is done either by trampling under cattle feet or bitten with stick.

6. Green gram (Moong) B.N. *Vigna radiata* L. wilczek Family :- *Leguminosae*

Economic Importance-

1. After gram, pigeon pea, Green gram is the third most important pulse crop in India.
2. It is a richest source of protein (25%) and supplies major share of the protein requirement of the population of the country.
3. It is extensively used as a split pulse as a *Dal* or consumed as a whole.
4. Green pods are used as a vegetable.
5. Green leaves and tops are feed to the animal or are utilized as a green manure.
6. Sprouted whole green gram is used for preparing *Usal* or curry.
7. It is easy to digest, hence preferred for the patient.
8. When green gram are allowed to sprout, ascorbic acid (vitamin C) is synthesized. The amount of riboflavin and thiamine are also increased.
9. Husk of pods or seeds with parts of kernel and leaves constitute a valuable cattle feed.
10. It is a leguminous crop, therefore possesses valuable properties of improving soil fertility.

Origin- According to Vavilov, India and Central Asia is native home.

Classification- Green gram in India is classified by Bose in 1932 who grouped them into 40 different types, mainly on the basis of characteristics variation. All these types have $2n = 24$. It has the following characteristics variation.

Classification on basis of-

Seed colour :- Seeds are Green, Black, Brown, or Yellow

Seed surface :- Dull or Shining

Flower colour :- light yellowish – olive or olive yellow

Pod colour :- Ripe pods are iron gray, olive gray or snuff brown

Botanical Description- It is a small herbaceous annual plant growing to a height of 30 to 100 cm. The central stems are more or less erect. While side branches are semi erect. The leaves are trifoliate with long petioles. Both stem and leaves are covered with short hairs. The flowers are of various shades of yellow colour and are produced in a cluster of 10 to 20 in auxiliary racemes on long pedicels. The pods are 6 to 10 centimeter long, round, slender with short pubescence. The colour of seed is usually green, but yellow, brown, or purple brown or black seeds also occur. The colour of cotyledons is yellow. The crop is fully self – fertile and self - pollinated

Area and Production in India-

State	Area (000ha)	Prod. (000 tones)	Productivity(q/ha)
Punjab	53.0	44.0	8.34
U. P	111.0	66.0	5.92
Orissa	777.0	396.0	5.09
Maharashtra	659.0	282.0	4.28
Tamil Nadu	150.0	63.0	4.20
Karnataka	177.0	71.0	3.99
Andhra Pradesh	434.0	166.0	3.82
Madhya Pradesh	131.0	43.0	3.25
Gujarat	174.0	47.0	2.69
Rajasthan	419.0	80.0	1.90
Total	3,299.0	1,374.0	4.17

Climate-Green gram is best suited to the area having an annual rainfall of 600 to 700 mm. Green gram is considered to be the hardiest of all pulse crop, because it is a drought resistant . It requires hot and humid climate.It is susceptible to frost and water logging condition. Heavy rains at flowering are harmful to the crop.

Soil :-A well drained medium to heavy soil is best suited . Saline and alkaline soils are not suitable for this crop.

Land preparation- Requires rough seed bed hence one deep ploughing (15 to 20 cm) followed by clod crushing and 2 to 3 harrowing are sufficient . Add organic matter at last harrowing @ 6-8 tons/ha.

Seed treatment-Seeds are treated with 1% mercurial compound or thiram or Captan or Dithane M – 45 as a preventive measure against seed borne diseases.*Trichoderma Viridae* @ 5 gm / Kg of seed as preventive measure against soil borne diseases.*Rhizobium* culture- inoculation of *Rhizobium* culture @ 250gm/ 10 Kg seed is recommended for enhancing activity of nitrifying bacteria for fixation of atmospheric nitrogen

Seed and Sowing- The Proper sowing time for Kharif is 15th June to 15th July. The second week of March to first week of April for the summer season.

- Spacing- 30 X 10 cm.
- Seed rate- 15 to 20 Kg /ha.
- Methods of Sowing- Drilling.
- Depth of Sowing- 3 to 4 cm

Manures and Fertilizers- Application of 6 to 8 tones of FYM or Compost at last harrowing. 25 Kg N/ha 50 Kg P₂O₅ /ha at the time of sowing

Interculture- Thinning Gap Filling- the gap filling should be done at 7 to 8 days after sowing. Thinning is carried out at 10 days after sowing, by keeping 1 or 2 seedlings per hill. Weeding and Hoeing :- Weed free condition at initial stage is very essential because its growth is very slow at this stage. 1 weeding and 1 hoeing at 20 to 22 DAS should be carried out.

Irrigation- For kharif season crop, irrigation is not needed but drainage is very important. There should be adequate drainage in the field because it is very sensitive to water logging condition. For Summer condition 5 irrigations should be given at pre sowing, 15, 30, 40, 50 DAS.

Insects and pest

1. Pod Borer- The adult lays eggs on leaves, the larva feeds on all plant parts and enter in the pod at pod setting and feed on the maturing grains. **Control measure-** Spray or dust any contact insecticide.

2. Hairy caterpillar- Adult lays eggs in mass. Larva feed on tender leaves may result in complete defoliation. **Control measure-** Manual picking of egg masses and caterpillars. Spray or dust any contact insecticide.

3. Galeuricid beetle- The adult beetle feed heavily on the flowers resulted in to flower drop. **Control measure-** Spray or dust any contact insecticide.

4. Leaf hopper, Jassids- Suck the cell sap from lower side of leaf. Can be controlled by spraying any systemic insecticide.

Diseases

1. Yellow mosaic- Yellow and Stunted plant growth. **Control measure-** The virus is spread due to whiteflies, jassids and Aphids. Control them by spraying systemic insecticide.

2. leaf curl- Leaves curled and retard the plant growth. **Control measure-** The virus is spread due to whiteflies, jassids and Aphids. Control them by spraying systemic insecticide.

3. Charcoal rot- Occasionally found. Plant dies due to dead tissues in stem. **Control measure-** Seed treatment.

4. Anthracnose- Lesions on leaves turn brown reduce metabolic activities of plant. **Control measure-** Spraying Copper oxychloride.

5. Cercospora leaf spot- Lesions on leaves turn brown reduce metabolic activities of plant. **Control measure-** Spraying of Copper oxychloride or carbendazim

6. Seed and seedling rot- Soil born fungal disease, After sowing seed rot or after emergence, the seedling shows wilting. **Control measure-** Make proper drainage, seed treatment with fungicide.

Crop rotation- Maize – Wheat – green gram, potato – Wheat – green gram, Wheat – green gram, green gram – potato.

Intercropping : Cotton + Green gram, Green gram + pigeon pea, Maize + green gram

Harvesting and Threshing-

The crop matures in 65 to 70 Days. Harvesting is done by picking pods 10 days interval, for 2 times. The crop is harvested before it is dead ripe. The pods are dried and threshing is done . At some places entire plants are uprooted or cut with sickle , dried and threshing is done by trampling or beating the produce with stick.

Yield :- 5 to 6 qt/ha

Varieties-

Name	Duration (Days)	Av. Yield q/ha	Season
Vaibhav	65-70	8-10	K
BPMR-145	65-70	8-10	K
Phule mung-2	65-70	9-9.5	K /S
Jalgaon 781	65-70	5-6	K
Kopergaon Mung.	60-65	5-6	K
P.S. 16	60-65	7-8	K
S-8	60-65	7-8	K
Pusa Baisakhi	60-65	6-7	S
D-41-9	60-65	4.5-5.5	K
M.L.-9	70-72	5-6	K
T-1	60-62	7-8	K
AKA-8802	65-70	8-10	K

7. Black Gram (Udid)B.N. *Vigna mungo* L. Hepper Family :- *Leguminosae*

Economic Importance-After Green gram, Black gram is the Fourth most important pulse crop in India. It is a richest source of protein (24%), Carbohydrates(60%), and fat (1.3 %). In Indian diet It is extensively used as a split pulse (*Dal*) or consumed as a whole as well as sprouted. It is a chief constituent of Indian *Papad*, *Halwa* and *imarti*, *Dosa* and *Idali*. It is used as a nutritive fodder specially for milking animal. It is also used as a Green manuring crop. Blackgram plant possess deep root system which binds soil particles and thus prevents soil erosion. It is the richest source of phosphoric acid than other pulses.

Area and Production in World-The crop is restricted to wet tropics (Semi arid region). It is cultivated in India, Pakistan, Sri-Lanka, Burma and up to some extent in south east Asia, Africa and America.

Area and Production in India-

State	Area(000 ha)	Prod. (000 ton)	Av. Grain yield (q/ha)
Karnataka	131	101	7.66
Orissa	504	321	6.37
Andhra Pradesh	525	284	5.40
Bihar	70	36	5.16
Gujarat	120	62	5.14
Maharashtra	501	230	4.59
Uttar Pradesh	278	124	4.44
West Bengal	77	32	4.09
Tamil Nadu	340	136	4.00
Madhya Pradesh	513	175	3.41
Total	3,293	1,596	4.85

Origin :- According to Vavilov, India is native of Black Gram

Classification :- by Bose in 1932 who grouped them into 2 sub species.

1. *Vigna mungo* V. *niger*- It include varieties which mature early and have bold seeds of black colour.
2. *Vigna mungo* V. *viridis* -It include varieties having Longer maturity period seeds are of small size and green colour.

Botanical Description :- It is a small herbaceous annual plant growing to a height of 30 to 100 cm. The stems is slightly ridged, covered with brown hairs and much branched from base. The leaves are large, trifoliate and hairy. The colour of the leaves are green to dark green. The inflorescence consist of a cluster of 5 to 6 flower. The pods are 4 to 6 centimeter long, cylindrical. There are 4 to 10 seeds in a pod. The seeds are generally black or very dark brown. The split seed of black gram is white in colour. The germination of seed is of epigeal type. Self fertilization – self pollinated crop.

Climate-Black gram is best suited to the area having an annual rainfall of 600 to 700 mm . The optimum temperature for better growth ranges between 25 °C to 35 °C but it can tolerate up to 42 °C because it is a hardy and drought resistant .It requires hot and humid climate .It is susceptible to frost and cloudy weather, Heavy rains at flowering are harmful for the crop.

Soil- A well drained medium to heavy soil is best suited . Saline and alkaline soils are not suitable for this crop.

Land preparation- Black gram requires fine leveled seed bed hence one deep ploughing (15 to 20 cm) followed by clod crushing and 2 to 3 harrowing are sufficient.Planking the field is the most essential operation for black gram cultivation.

Seed and Sowing- It is cultivated in Kharif and summer season. The optimum sowing time for Kharif is 15th June to 15th July.The third week of February to first week of April for the summer season.

- Methods of Sowing :- Drilling
- Depth of Sowing :- 3 to 4 cm
- Spacing- 30 X 10 cm
- Seed rate- 15 to 20 Kg /ha

Seed treatment-Seeds are treated with 1% mercurial compound or thiram or Captan or Dithane M – 45 as a preventive measure against seed borne diseases.*Trichoderma Viridae* @ 5 gm/Kg of seed as a preventive measure against soil borne diseases. Rhizobium culture- inoculation of Rhizobium culture @ 250gm/ 10 Kg seed is recommended for enhancing activity of nitrifying bacteria for fixation of atmospheric nitrogen

Manures and Fertilizers- Application of 6 to 8 tones of FYM or Compost at last harrowing .25 Kg N/ha 50 Kg P₂O₅ /ha at the time of sowing

Interculture- Gap Filling :- the gap filling should be done at 7 to 8 DAS.Thinning :- thinning at 10 DAS, by keeping 1 or 2 seedlings.Weeding and Hoeing :- Weed free condition at initial stage. 1 weeding and 1 hoeing up to 40 DAS.Chemical weed control :- pre – emergence spray of Basalin @ 1.0Kg a.i./ha.

Irrigation- for kharif season crop, irrigation is not needed but drainage is very important. There should be adequate drainage in the field because it is very sensitive to water logging condition. For Summer condition 5 irrigations should be given on following DAS Pre sowing irrigation, 15 DAS 30DAS 40 DAS 50 DAS.

Crop rotation- Maize – potato – Black gram, Maize - Toria – Black gram, Maize - Wheat – Black gram, Paddy – Wheat - Black gram, Black gram – Wheat - Black gram, Black gram – Onion

Intercropping :Cotton + black gram, Black gram + pigeon pea, Maize + black gram.

Insects and pest-Hairy caterpillar, Leaf hopper, Jassids

Diseases :-Yellow mosaic, leaf curl, Charcoal rot, Anthracnose, Cercospora leaf spot, Seed and seedling rot.

Harvesting and Threshing- The crop matures in 75 to 80 Days. Harvesting is done by cutting with sickle. Dried and threshing is done by trampling or beating the produce with stick.

Yield - 8 to 10 qt/ha.

Improved Varieties-

Name	Duration (Days)	Av. Yield (Q/ha)
TPU- 4	70-75	11-12
TAU- 1	70-75	10-11
BDU-1	75-80	8-10
Shindkheda	75-80	10-11
T-9	70-80	10-12
D-6-7	70-80	10-11
No.55	70-80	10-12
Krishna	70-80	10-12

8. Pigeon Pea (Tur, Red gram, Arhar) *B.N. Cajanus cajan* L. Millsp

Family :-*Leguminosae*

Economic Importance-

1. After gram, pigeon pea is the second most important pulse crop in India.
2. Mostly grown for grains, green manuring and fodder.
3. It is a rich source of protein (22.3%) and supplies major share of the protein requirement of the population of the country
4. The seed of pigeon pea are rich in iron and iodine
5. It is extensively used as a split pulse as a *Dal*
6. Green pods are used as a vegetable
7. Husk of pods or seeds with parts of kernel and leaves constitute a valuable cattle feed.
8. Dry sticks are used as a fuel or basket making.
9. It is a leguminous crop, and has valuable properties of improving soil fertility

Origin-It seems that Africa is the probable place of origin and the plant might have been introduced in India by ancient traders. According to Vavilov, India is native home but has no reference to this thought.

Classification :- The cultivated type of *Cajanus* genus are classified into two groups

1. *Cajanus indicus* var. *bicolor* – this group includes generally late maturing varieties, having tall, bushy plants, flower yellow, purple streaked, pods dark colour each with 4-5 seeds
2. *Cajanus indicus* Var. *Flavus* – This group includes early maturing varieties, having smaller plants, yellow flower, plain pods with 2 to 3 seeds.

Area and production in world-India rank first with about 90% of area and 85% of Production. It is also grown in Malawi, Uganda, Nepal, Myanmar and in Dominican republic of America

Area and production in India-It is grown on 3.61 million ha. Area with total production of about 2.70 million ton with average yield of 7.47 q/ha. Maharashtra, Haryana, U.P., M.P., Gujarat and Karnataka are the major Pigeon pea growing state in India.

Botanical Description-It is an erect shrub. The height ranging from 1 to 4 meter tall. **Root** – The root system of pigeon pea consists of a central tap root with numerous lateral secondary branches. **Leaves** – leaves are trifoliate compound, central leaflet longer than lateral ones. **Inflorescence** – The inflorescence is an axillary raceme often forming a terminal panicle. The flowers are distinctly papilionaceous. The process of flowering continues in each plant almost up to the time of harvest. The flowers are self-pollinated, pollination takes place before the flower opens. Cross pollination may also occur to some extent. **Pods** – The fruit of pigeon pea is a pod. These vary in length, width and nature of markings. The length varies from 5 to 10 cm, width from 0.6 to 0.9 cm. The pods

vary in colour from green to dark brown. The seeds within the pods vary in number, but there are usually 4 to 5 in each pod in late maturing varieties and 2 to 3 in early maturing varieties. **Seeds** – Seeds differ a great deal in size, shape and colour. Seeds are round or lens shaped, the colour of the seed coat being dirty white to silver white, light brown to chestnut brown, dark mottled brown and pinkish black and the cotyledons of yellow colour.

Climate-Pigeon pea grows well in warm tropical and subtropical climate. The crop prefers a fairly moist and warm climate during its vegetative growth, and bright sunny weather, during the flowering for the setting of fruits. It is highly susceptible to frost at the time of flowering. Cloudy weather and excessive rainfall at flowering time damage the crop to a great extent.

Soil- Pigeon pea can be grown on wide range of soils varying from sandy loam to clay loam. It performs well on fertile and well drained loamy soils. The saline, alkaline and water logging soils are undesirable. It is successfully grown in black cotton soils having good drainage facilities.

Land preparation-Pigeon pea requires fine and compact seed bed hence one deep ploughing (18 to 20 cm) followed by clod crushing and 2 to 3 harrowing are sufficient.

Seed and Sowing-The optimum sowing time for pigeon pea is First fortnight of June.

- Methods of Sowing :- Drilling , Dibbling.
 - Depth of Sowing :- 4 to 5 cm
 - Spacing- For Early varieties – 45 X 20 cm
For mid late varieties – 60 X 30 cm
For late varieties – 75 X 20 - 25 cm
- Seed rate- For Early varieties - 18 to 20 Kg / ha
For midlate varieties – 15 Kg / ha
For late varieties – 12 Kg / ha

Seed treatment-Seeds are treated with 1% mercurial compound or thiram or captan or Dithane M – 45 as a preventive measure against seed borne diseases. Trichoderma Viridae @ 5 gm / Kg of seed preventive measure against soil borne diseases. Rhizobium culture- inoculation of Rhizobium culture @ 250gm/ 10 Kg seed is recommended for enhancing activity of bacteria for fixation of atmospheric nitrogen

Fertilizers and manures-Application of 6 tones of FYM or Compost at last harrowing. 25 Kg N + 50 Kg P₂O₅ / ha at the time of sowing

Interculturing :- Thinning- The thinning is very important in pigeon pea thinning was carried out at 15 days after sowing, by keeping 1 or 2 seedlings per hill. Weeding and Hoeing :- Weed free condition at initial stage up to 60 days is very essential. 2 weeding and 2 hoeing at 25 DAS and at 30DAS should be carried out.

Crop rotation- Pigeon pea – Wheat, Pigeon pea – late potato, Pigeon pea – lentil, Pigeon pea – Sugarcane, Pigeon pea – Wheat – green gram.

Intercropping-Pearl millet + Pigeon pea (2:1), Sunflower + Pigeon pea (2 : 2) , Soybean + Pigeon pea (4:1).

Insect pests-

1. **Pod Borer**-The larva feed on almost all plant parts, and bore in to pod at pod formation stage and eat away all grains.**Control measures**- Integrated pest management- clean cultivation, Use of Pheromone traps, Spray neem seed extract, spray of HaNPV (Heliokill), Spray insecticide.
2. **Hairy Caterpillar**- The larva feeds on tender leaves, resulting in complete defoliation of plant.**Control measures**-Manual picking of egg masses and caterpillars and destroying them, dusting with insecticide powder or spraying insecticide.
3. **Pod Fly**- The adult flies lay eggs on the pod, on hatching small larvae bore in to pod and eat on developing grains. Afterwards pupates inside and adult fly come out after maturity by making big hole on the pod.**Control measure**- Serious pest in North India, Spraying the systemic insecticide like Dimethoate at early pod formation stage to prevent and control the pest.
4. **Bristle beetles**- The adult beetle feed heavily on flower and bring flower drop.**Control measure**- Hand picking of adults, dusting insecticide.

Diseases-

1. **Wilt**- The affected plant show yellowing and drying of leaves, followed by drying of entire plant. Roots and base of stem show black or white streaks.**Control measure**-Remove and destroying of affected plants, follow mix or inter cropping, crop rotation and growing resistant varieties.
2. **Sterility mosaic virus**-Stunted growth, Bushy and pale green leaves with ring spots, and lack of reproductive parts.Small insects like mites transmit this disease.**Control measures**-Growing a resistant varieties, removing and destroying of affected plants. Spraying of Metasystox.
3. **Phytophthora Blight**- Sudden wilting of plant parts, upward rolling of leaves, withering of petioles, dark brown lesions on stem, petioles and leaves. **Control Measures**-Growing resistant varieties, Removing affected plants, avoiding water logging
4. **Leaf Spot**-Gray colour angular lesions on leaves.**Control measures**-Seed treatment, spraying of Zineb 0.2 %.

Harvesting and Threshing-

Harvesting is done two third to three – fourth of pods turn brown. Or when most of the leaves have dried up and shed. Pods may be picked at frequent intervals or finally the plants are cut close to the ground, tied into bundles carried out in threshing yard stocked upright, dried for 8 to 10 days. Threshing should be done by beating the pods with stick or trampled by bullock. The seed and chaff are separated by winnowing .

Varieties-

Name	Duration	Av. Yield (q/ha)	Characters
ICPL- 87	125-130	18-20	Early, Pods in cluster, suitable for rotation.
Vipula	145-160	24-26	Resistant to wilt and sterility mosaic , medium size red seeds.
BSMR-736	160-170	18-20	Resistant to wilt and sterility mosaic , Bold size red seeds.
BSMR-853	160-170	18-20	Resistant to wilt and sterility mosaic , medium size white seeds.
AKT- 8811	160-170	15-16	Medium size red seeds, release for Vidharba region.
Amol	160-170	20-22	Medium size red seeds, release for Marathwada region.
Rajeshwari	140-150	28-30	Resistant to wilt & sterility mosaic, mid late maturing, red colour bold grain size, released for Maharashtra state

9. Cow pea (Chawali), B.N. *Vigna sinensis* L. , Family – Leguminosae

Economic Importance-

1. It is used as a Pulse, a fodder, and a green manure crop.
2. Cowpea grain contains 23.4 % protein, 1.8 % fat, 60.3 % carbohydrates, Calcium and iron therefore, It is called as Vegetable Meat.
3. Grain is also used as a concentrate feed for cattle, and stalk forms excellent forage.
4. The fodder value of cowpea stalk is quite high and comparable with to Lucerne.
5. It is also grown as to check the soil erosion

Origin-Cowpea is probably a native of central Africa where almost all wild forms are found.

Classification- According to Piper all the cultivated types have been classified in to three groups

1. *Vigna sinensis* (cowpea) - Pods are 20-30 cm long containing 0.6 to 0.9 cm long seeds. On drying pods are neither flabby nor inflated.
2. *Vigna sinensis* sub sp. *Catjang* (Indian cowpea)- Pods are 7.5 to 12.5 cm long, erect or bending when green, spreading or deflexed when dry. Seeds are 0.5 to 0.6 cm long and nearly as thick as broad .
3. *Vigna sinensis* sub sp. *Sesquipedalis* :-Pods are 30-60 cm long and pendent. On drying pods are inflated flabby and shrinking between seed before drying. Seeds are 0.8 to 1.2 cm long.

Botanical Description- The cowpea is a twining annual herbaceous plant. The root system consist of welldevelop tap root system.The stem is slightly ridged and almost glabrous.The leaves are trifoliate, alternateand with scattered short hairs.The flower are white, yellow, or pink in colour and are usually self pollinated. Pods are long and cylindrical and constricted between seeds. The seeds are bean-shaped and many times spotted with different coloures such as brown, green yellow, white and mottled.

Climate-It can be grown in all tropical and sub tropical climates. Being a warm weather crop it can withstand in drought condition up to a certain limit.Germination of cowpea is better between the temperatures 12°C and 15 °C, the crop thrives best in the temperature 27°C and 35 °C . It can be grown under shade of trees but can not tolerate cold or frost

Soil-It can be grown on a variety of soils. Well drained loam or slightly heavy soils are best suited for cowpea crop. Saline or alkaline soils are not suited to cowpea cultivation.

Land preparation- Cow pea requires smooth seed bed hence one deep ploughing (18 to 20 cm) followed by clod crushing and 2 to 3 harrowing are sufficient. Planking the field is the most essential operation for cow pea cultivation.

Seed and Sowing-The proper sowing time for *kharif* cow pea is second fortnight of June to first week of July, and March to April for the summer season. In hilly areas, this crop is sown in April-May for green manuring

Methods of Sowing- Drilling
Depth of Sowing- 4 to 5 cm
Spacing :- a) 30 X 10 cmb) 45 X 10 cm
Seed rate : For grain or vegetable purpose 20 to 25 Kg /ha. For fodder purpose 35 to 45 Kg /ha

Seed treatment-Seeds are treated with 1% mercurial compound of thiram or Captan or Dithane M – 45 as a preventive measure against seed borne diseases.*Trichoderma Viridae* @ 5 gm / Kg of seed as preventive measure against soil borne diseases.Rhizobium culture :- inoculation of Rhizobium culture @ 250gm/ 10 Kg seed is recommended for enhancing activity of bacteria for fixation of atmospheric nitrogen.

Fertilizers and manures- Application of 5 to 6 tones of FYM or Compost at last harrowing . 25 Kg N/ha + 50 Kg P₂O₅ /ha + 25 Kg K₂O at the time of sowing

Interculturing :-Gap Filling- 5 to 7 days after sowing.Thinning - 10 days after sowing, by keeping 1 or 2 seedlings. Weeding and Hoeing-Weed free condition at 20-22 DAS is very essential because its growth is very slow at this stage . 2 weeding and 2 hoeing up to 50 Days should be carried out. Chemical weed control- pre-emergence application of Fluchloralin @ 1.0Kg a.i./ha in 800 to 1000 liter of water is effective for controlling weeds.

Irrigation- For *kharif* crop, irrigation is not needed, but in *Summer* 3 irrigations should be given at Pre sowing , Flowering and Pod filling stage.

Crop rotation-

For grain or vegetable purposes

- Maize-Wheat – cowpea, paddy-Wheat – cowpea,
- Maize-potato – cowpea, Maize-toria-Wheat –cowpea

For fodder purpose

- Jowar+ cowpea- Berseem-maize + cowpea
- Maize – oat-maize + cowpea
- Maize - Berseem- maize + cowpea
- Jowar- Berseem-maize + cowpea

Insect and pest-

1. Hairy caterpillar- feed heavily on vegetative parts collection and destruction of egg and larva masses and dusting of contact insecticide dust will be sufficient.

2. Leaf Hopper- Jassids and Aphids- Small insects suck the sap and reduce the plant growth. Spraying of systemic insecticide is advised.

3. Cutworms- The larvae cut the plant at early age during night.

4. Pod Borer- Larvae bore in to pod and feed on developing grains.

To control this pest deep ploughing and spraying any contact insecticide at flowering and pod formation stage is recommended.

Nematodes-Plant parasitic species damage the root by making lesion, galls or root knots. Application of neem cake at sowing is only control in Indian conditions.

Diseases-

1. **Bacterial blight of cowpea**- seed born disease, the germinating seedling turn brown-red and die. Use of healthy seed and seed treatment are the easy control measures.
2. **Cowpea mosaic**- this is a viral disease through vectors (aphids and white flies). Affected plant shows chlorotic area on leaves. Remove and destroy the affected plant, Spray systemic insecticide to control vectors.
3. **Powdery Mildew**- The affected plant show powdery spots on leaves and stem turns yellow. Grow resistant varieties, Dusting of Sulphur @25kg/ha or spraying wettable sulphur @2-4 kg/ha
4. **Rust**- Circular reddish brown pustules on leaves, stem and pods. Control measures- Destroy the affected plants, Spray Zineb for three times with 10 days interval.
5. **Wilt**- Young plant start drooping green leaves, stem cut show brown discoloration. Seed treatment with Trichoderma Viridae @ 5 gm Kg and Captan or Thiram @2.5g/kg.

Varieties-

Name	Duration	Av. Yield(q/ha)
C-152	105-110	15-20
Cowpea -74	110-120	15-20
Pusa sawani	110-120	15-20
V-240	110-120	15-20
Konkan Sadabahar	60-65	12-15
Konkan safed	70-75	14-16
Phule Pandhari	70-75	14-16

10. Horse Gram (Kulthi) B. N. *Dolichos biflorus* (Lam) Verde, *Macrotyloma uniflorum* (Lam) Verde , Family :- papilionaceae

Economic Importance-

1. It is used as a Pulse, fodder, and green manure crop.
2. It contains 22 % protein, 1 % fat , 62 % carbohydrates
3. It is also rich source of Calcium 3.1 (mg) and iron 90 mg.
5. It is mainly consumed on large scale in the south India.

Origin- South East Asia is native home of horse gram

Botanical Description :-It is a small herbaceous sub erect, annual plant growing to a height of 30 to 40 cm.Horse gram posses a slightly angular, tough hairy stem,Branches and petioles are purplish.Leaves are alternate, pulvinate and innately trifoliate.Flowers vary in colour , but are mostly purple. They are bisexual and self pollinated.Pods are linear dehiscent and borne on leaf axils.Seeds are either lighter shades of red, brown or black.The crop matures between 80 and 125 days depending on the genotype

Climate- Horse gram has excellent adaptability to drought and harsher environments prevailing in the semi arid . It grows well under scanty rainfall conditions receiving less than 750 mm rainfall annually.

Soil- Horse gram is grown on a wide range of soils such as sandy , loamy or even deep vertisols. However, it is often the first crop sown into cultivation on marginal lateritic soils

Land preparation :-Horse gram requires rough seed bed hence one deep ploughing (15 to 20 cm) followed by clod crushing and 2 to 3 harrowing are sufficient.

Seed and Sowing- In South India it is cultivated in Kharif season The optimum sowing time for Kharif is late August to November.

Methods of Sowing :- Broadcasting , Drilling

Depth of Sowing :- 3 to 4 cm.

Spacing :- 30 X 10 cm

Seed rate :- 20 to 25 Kg / ha for grain purpose, 40 Kg /ha for Fodder purpose

Seed treatment-Seeds are treated with mercurial compound of thiram or Ceresan 2-3 g/Kg as a preventive measure against seed borne diseases.Trichoderma Viridae @ 5 gm / Kg of seed as a preventive measure against soil borne diseases .Rhizobium culture :- inoculation of Rhizobium culture @ 250gm/ 10 Kg seed is recommended foe enhancing activity of nitrifying bacteria for fixation of atmospheric nitrogen.

Fertilizers- 20 Kg N + 40 Kg P₂O₅ /ha at the time of sowing.

Interculture- Gap Filling :- the gap filling was carried out at 7 to 8 days after sowing.Thinning :- The thinning is carried out at 10 DAS, by keeping 1 or 2 seedlings per hill.Weeding and Hoeing :-

Weed free condition at initial stage is very essential because its growth is very slow at this stage . 1 weeding and 1 hoeing at 20 to 22 DAS should be carried out.

Harvesting and Threshing- At Maturity pods are slightly brittle and straw colour becomes yellow. The plants are uprooted , stacked for few days to dry and threshing is done by trampling or beating the produce with stick.

Yield :- 7 to 8 q/ha

Insect and pest-Hairy caterpillar, pod borer, Leaf eating caterpillar

Diseases-Root rot, Leaf spot, Die back, Anthracnose, Bacterial leaf spot, Leaf rust

Varieties	Grain yield q/ha
Sina	7-8
Man	6-7
CVDB-6	8.6
K-42	7.2
BGM-1	5.9
HPK-2	8-10
VZM-2	8-10
PGH-9	4.7
VLG-4	6-8
PHG-62	12 -13
Phule Sakas	8 – 9

11. Moth bean (Dew bean, Mataki), B.N. *Phaseolus aconitifolius*, Family – Leguminosae

Economic importance-

1. It is an important pulse crop of the desert region.
2. It contains 25 % protein.
3. It is also grown for fodder purpose.
4. It enhances the soil fertility by fixing the atmospheric N.
5. It is also used as a green manuring crop.
6. It has highest power to tolerate the drought in comparison to other pulse crop of arid and semi arid region.

Origin- India is the native of moth bean.

Climate :- Moth bean is best suited to the area having an annual rainfall of 300 to 700 mm. Moth bean is considered to be the hardiest in all pulse crop, because it is a drought resistant. It requires warm climate. It is susceptible to frost and water logging condition of the soil. Heavy rains at flowering are harmful to the crop.

Soil- It can be grown on light, alluvial to heavy loam soils. Saline and alkaline soils are not suitable for moth bean cultivation. Soils with neutral pH are best suited for moth bean cultivation.

Seed and sowing :- In Rajasthan, Haryana and Gujarat it is cultivated in *Kharif* and *summer* season. In Maharashtra and Uttar Pradesh it is cultivated only in *Kharif* season. The optimum sowing time for *Kharif* is 15th June to 15th July. The second week of February to first week of April for the *summer* season.

Seed rate-

Pure crop –	12-15 Kg/ha
Mixed crop-	4-5 Kg/ha
Fodder –	20-25 Kg/ha
Spacing :-	30 X 10, 45 X 10 cm

Seed treatment- Seeds are treated with 1% mercurial compound or thiram or Captan or Dithane M – 45 as a preventive measure against seed borne diseases. *Trichoderma Viridae* @ 5 gm / Kg preventive measure against soil borne diseases. **Rhizobium culture :-** inoculation of *Rhizobium* culture @ 250gm/ 10 Kg seed is recommended for enhancing activity of nitrifying bacteria for fixation of atmospheric nitrogen

Soil treatment- To control the white grub, 25 Kg/ha of Phorate 10 G should be applied. If the termite problem is serious, apply 5% Folidol dust @ 25Kg/ha at the time of land preparation.

Fertilizers and manures :- Application of 6 to 8 tones of FYM or Compost at last harrowing. 25 Kg N + 50 Kg P₂O₅ / ha at the time of sowing.

Interculture- Gap Filling :- the gap filling should be carried out at 7 to 8 days after sowing. Thinning :- The thinning is very important in Moth bean thinning should be carried out at 10 DAS, by keeping 1 or 2 seedlings per hill.

Crop rotations under dry condition: Sorghum-Moth bean-barley, Moth bean – pearl millet – Mustard, Moth bean – gram, Maize – moth bean- Mustard, Moth bean –Linseed,

Under Irrigated condition : Moth bean- potato- wheat, Moth bean- radish – wheat, Moth bean – Toria – potato, Moth bean – Toria- wheat- Green gram.

Harvesting and threshing –The crop matures in about 90-100 days . The main sign of maturity of moth bean is that the pods become yellow black.The crop may be harvested with sickles and then dried and threshed.

Yield :-A good crop may yields about 6-8 quintal grains and 8-10 quintals straw/ha

Insects and pest - Hairy caterpillar, Leaf hopper, Jassids

Diseases-Yellow mosaic, leaf curl, Charcoal rot, Anthracnose, Cercospora leaf spot, Seedling rot

Varieties

1. Jadia (Rajasthan)
2. Baleswar-12 (Gujrat)
3. Mevi (Gujrat)
4. RMO-40 (Uttar Pradesh)
5. T-3 (Uttar Pradesh)
6. Nadisd 8-3-2 (Gujrat)
7. Jagudam 9-2 (Gujrat)
8. Jawel 12-1 (Maharashtra)
9. Gulia 3-5 (Maharashtra)
10. MBS-27 (Maharashtra)
11. Types 4301,4312,4312 (Uttar Pradesh)

12. Groundnut, B. N. *Arachis hypogaea*, Family :-*Leguminosae*

Economic importance-

- Among the Oilseed crops, groundnut has first place in India. It is also known as peanut, earthnut, monkey nut, goober, pinda, and manilla nut.
- G.nut seed contains about 45 % oil and 26 % protein. Oil is primarily used in the manufacture of vegetable oil (vanaspati ghee).
- G.nut kernel is highly digestible, the raw seeds are consumed either by roasting or boiling the pods.
- The biological value of G.nut protein is among the highest of the vegetable protein, equal to that of casein
- G.nut seed is a good source of all vitamins except B₁₂
- They are a rich source of thiamin, riboflavin, nicotinic acid, and vitamin E. However they lack in vitamin A.
- The minerals like phosphorus, calcium, and iron are present in significant amount.
- The oil cake obtained from the extraction of the oil is a valuable organic manure and animal feed. It contains 7 - 8 %N + 1.5% P₂O₅ + 1.5 % K₂O.
- It is a good rotation crop it build up the soil fertility by fixing atmospheric nitrogen through the root nodules and also an efficient cover crop for lands exposed to soil erosion

Origin- De Gandolle in 1825, stated that groundnut is originated in Brazil in south America, the plant was introduced In India by Portuguese.

Classification-According to Waldron Gr.nut have been divided in two groups

- *Arachis hypogaea* - The bunch or erect type the sub species *fastigiata*.
- *Arachis hypogaea* - The trailing or spreading type the sub species *proclumbens*

According to Specialist of Tamil Nadu state :- the cultivated *A. hypogaea* has been classified into five groups based on habit of growth, testa colour, size and other character of pods, and leaf.

- 1) *Arachis hypogaea* var. *oieifera*
- 2) *Arachis hypogaea* var. *nambyquare*
- 3) *Arachis hypogaea* var. *rasterio*
- 4) *Arachis hypogaea* var. *asiatica*
- 5) *Arachis hypogaea* var. *gignatea*

Area and Production in world-

- During 1997 about 22.48 million ha. Area was under Groundnut crop with about 29.19 million tons of production.
- India rank first in area, China rank first in total production and SA rank first in productivity of Groundnut.
- Nigeria, Indonesia, Senegal, Zaire, Myanmar, Argentina and Sudan are the important Groundnut growing countries.

Area and Production in India-

- In India it was grown on 8.26 million ha. (1995) with the total production of about 7.81 million tons.

- 85% of total area is under *kharif* crop grown over all the country, 10% area under the *Rabi* Groundnut grown in southern region in rice fallows and 5% area under *Summer* crop grown in central and north-western part of the country.
- Andhra Pradesh, Gujarat, Tamil Nadu, Karnataka and Maharashtra are major Groundnut producers.
- In Maharashtra it is grown on 5.80 lakh ha. With total production of 7.30 lakh tons during 1995.

The area and production of *kharif* Groundnut has been shifted to Soybean and other crops and the area under Summer Groundnut increased due to following reasons. Erratic rains i.e. late onset (late sowing), long dry spells at flowering or pegging and heavy late rains at harvesting, Labour shortage and low yields in *kharif*.

Botanical description-G.nut. Plant possesses tap root system. Stem of matured plant is cylindrical. It has large pith which in the older portion becomes hollow. The lateral arising from stem are called as branches. Leaves are borne spirally on the main stem. The leaves consist of 2 pair of leaflets which are opposite with entire margin. The flower arises in leaf axis in single or in cluster of 3 or more having yellow colour. The young flower opens before 6 a.m. and fertilization is completed before midday. The basal portion of ovary starts to elongate and peg is visible after 5 to 7 days. In bunch type pegs enter in soil within 2 to 8 days. The purple color pegs are having about 2 to 6 cm length. The pod formation begins after 5 to 6 days of entrance of pegs into soil. The depth of pods depends upon the length of pegs. Complete development of groundnut pods takes place up to 60 days after flowering. Seed is composed of two large cotyledons.

Climate-Gr.nut is cultivated in tropical and sub tropical countries. Gr.nut is mostly grown as rain fed crop, during *kharif*. This crop does not tolerate frost, continuous heavy rainfall as well as severe drought or water logging condition. The optimum well distributed rainfall of about 600 to 750 mm is necessary for this crop. Gr.nut thrives best at temperature range of 21⁰c to 26.5⁰c. Low temperature at initial stage is harmful. The long dry spell during flowering and peg formation, reduces the pod yield. The rainfall at harvesting stage is harmful. This crop needs adequate moisture at germination, flowering, pegging, and pod development stages for better crop yield.

Soil-Gr.nut is cultivated in all types of soil. Sandy and sandy loam soils as well as light soil helps in easy penetration of pegs and their development. Heavy and stiff clay soils are not suitable for Gr.nut cultivation, as they become hard during dry weather and hinders the pod formation due to which difficulties can arise in harvesting. The pH range below 5 and above 8 is unfavorable for Gr.nut crop. The optimum pH range should be 6 to 7.

Preparation of Land – Gr.nut crop requires a deep, loose, friable and fine seed bed. One ploughing with MB plough followed by clod crushing and 2 to 3 harrowing would be sufficient to bring the fine tilth of the field.

Seed and Sowing :-Time of sowing- *Kharif* Gr.nut is sown immediately after onset of monsoon (15th June to 15th July) the delayed sowing of Gr.nut crop beyond 15th July decreases the yield considerably.

Seed Treatment-Healthy ,well developed bold seed should be selected. Seeds are treated with 1% mercurial compound or thiram or captan or Dithane M – 45 as a preventive measure against seed borne diseases like root rot, bud necrosis and wilt. *Trichoderma Viridae* @ 5 gm/Kg preventive measure against soil borne diseases. *Rhizobium* culture :- inoculation of *Rhizobium* culture @ 250gm/ 10 Kg seed is recommended for enhancing activity of nitrifying bacteria for fixation of atmospheric nitrogen

Methods of Sowing- Drilling , Dibbling

Depth of Sowing :- 4 to 5 cm

Spacing and seed rate :-

Type	Spacing (cm)	Seed rate (kg/ha)
Bunch or Erect type	30x 10,15	100-110
Semi sprayding	37.5x 15	90-100
Sprayding	45 x 15, 20	75 – 90

Manures and Fertilizers- 5 to 7.5 tones of FYM/ Compost before last harrowing. Under assured rainfall condition & for irrigated condition- 25 kg. N and 50 kg. P_2O_5 /ha .For Rain fed & Dry land - 20 kg.N and 40 kg. P_2O_5 /ha. During Summer- 25 kg.N and 50 kg. P_2O_5 /ha. It is recommended to apply 250 Kg Gypsum /ha for better pod filling (Half at sowing + Half at 30 DAS)

Use of Micronutrients-If the soil is deficient in Boron- Boron spray @ 0.1PPM (0.3g boric acid or 0.5 g borax in 500 liter of water /ha at 30 and 50 days of the crop age. If the soil is deficient in Fe- soak 2.5 to 3 Kg ferrous sulphate in 10 liter of water for night and prepare the solution by dissolving 2.5 Kg Urea in 500 liter of water and spray on the crop of gr. Nut. If the soil is deficient in Zn- Add 30 Kg $ZnSO_4$ in soil once in three years.

After care-Gap filling should be completed within 8 – 10 days from sowing. 2 – 3 hoeing and 1 to 2 weeding should be done depending upon weed intensity, before peg formation (45DAS). Chemical weed control- pre – plant incorporation of Fluchloralin or @ 1.0Kg a.i./ha in 800 to 1000 liter of water is effective for controlling weeds. Earthing up and Rolling of empty drum are the important Interculture operations followed in Gr.nut

Irrigation Management –Gr, nut crop requires an average of 40 to 45 cm/ha of water. If there is long dry spell in monsoon then 2 to 3 irrigations at Flowering, Peg Formation and pod development stage. *During summer season most of Groundnut crop is cultivated with Polythin mulch.*

Crop rotation are followed in gr.nut-Jowar – Gr.nut , cotton -- Gr.nut,cotton-- Gr.nut -- jowar, Gr.nut – Tobacco– jowar, Gr.nut -- Wheat

Harvesting-The erect, sprayding and semi- sprayding type of varieties become ready for harvest at different times. The sign of Maturity of Gr.nut are as - Vine turn to yellow, Leaves start shedding, When the pods shacked in hand a specific sound is observed, When the pod is opened a blackish spot is observed at inner side of the shell, If kernel pressed in teeth it will break down into pieces . The bunch type of varieties are harvested by pulling the plant. The spreading and semi- spreading type of varieties are harvested by spade, harrow, local plough,or groundnut digger.The pods are separated from the plant with hand

Varieties-

Name	Season	Duration	Av. Yield (q/ha)
Erect Types			
SB-11	K/S	105-110, 115-120	12-14, 20-25
JL- 24 (Phule Pragati)	K	90-95	18-20
TAG-24	K/S	100-105, 110-115	12-14, 25-30
JL-220 (Phule Vyas)	K	90-95	20-24
JL-286 (Phule Unap)	K/S	90-95, 110-115	16-18, 20-24
TPG-41	R/S	125-130	25-28
TG-26	K/S	95-110, 110-115	15-16, 25-30
JL-501	K/S	99-104, 110-115	16-18, 25-28
Semi- Spreading Types-			
TMV-10	K	125-130	20-25
Spreading Types-			
M-13	K	130-135	20-25
Karad-4-11	K	130-135	20-25

Insect Pests-

1. White Grub- The adult larva living under ground, damage the crop by eating roots. The patches of dry plants are observed in the field. **Control Measure-** Use only well decomposed FYM/Compost.Deep ploughing in summer to expose insect to sunlight and Birds.Destroy the Adult Grubs by collecting them in early night from their shelters i.e. Babul and Neem trees.Apply contact insecticide in powder form in the soil at early stage of larva (First week of June).

2. Leaf minor-The dark headed green or brown larvae mine in to tender leaves, which look like blister mine. The attacked plant do not grow properly. Insect remain active from July to December. **Control measures-**Spray Metasystox or Dimecron.

3. Aphids-Adult and nymphs prefer to feed on young shoots, causing leaf curl, and stunted growth.**Control measures-**Spray Metasystox or Dimecron.

4. Termites-Feed on roots with the result that the plant wither. The attack continues on pods. **Control measures**-Apply 5% Folidol dust in soil @ 25 kg/ha

Diseases-

1. Tikka- (*Cercospora*) -Small dark brown circular spots appear on the leaves. The disease sprayed fast at high humidity and temp. Severe attack may result in to complete defoliation.**Control measures**-Seed treatment with Thiram

2. Sclerotium Rot-It is soil born, Affected plant parts show the development of white thread like fungal growth near the soil surface.**Control measures**-Collect and burn the affected plants, seed should be treated with Trichoderma Viridae @ 5 gm Kg

3. Rosette- (Bud Necrosis)-Viral disease transmitted through Aphids. Stunted and bushy plant appearance, reduction in size of leaflets. **Control measure**-Control the vector, and remove the affected plants.

4. Rust-Development of red pustules on leaves, usually on lower surface.**Control measures**-Destroy the affected plants, Spray Zineb for three times with 10 days interval.

12. Soybean, *B.N. Glycine max* L. Merril. Family :- Leguminosae

Area and Production in world(m ha) - Area in 2010-11- world-117.5, USA-41.10, Brazil- 32.9, Argentina-24.70, China-8.22, India- 4.70, Paraguay- 3.52, Canada- 1.17 and others- 1.19

Economic Importance-

- As like a milk, Soybean grains are complete food, because It contains 40% high quality protein, 20% oil, rich in the valuable amino acid lysine (5%), minerals, salt and vitamins (Thiamine and Riboflavin), and when sprouted considerable amount of vitamin C and Vitamin A.
- As a human food it is used in the form of chapati, milk, sweets, pastries.
- Soybean is used for the manufacturing soap, paints, plastics, Linoleum, Candles, oil, cloth varnish, enamels, rubber, printing ink and lubricants.
- Soybean is used for the preparing Vanaspati ghee
- Soybean oilcake is a nutritive food for live stock and poultry.
- Soybean builds up the soil fertility by fixing large amounts of atmospheric nitrogen through root nodules , and also through leaf fall on the ground at maturity.
- Soybean being the richest ,cheapest and easiest source of best quality proteins and fats and having a vast multiplicity of uses as food and industrial products is called a wonder crop

Origin- Soybean is a native of Asia- China. The wild forms of soybean observed in China. This crop is introduced in India during 1963.

Classification-

1. **Manchurian Classification** :- this classification is based on colour of seed. Soybean have been divided into three groups according to this classification

A. Yellow group :-

- i. Yellow seed with light hilum
- ii. Yellow seed with Golden hilum
- iii Yellow seed with Brown hilum

B. Black group-

- i. Large black seed
- ii Flat black seed
- iii Small black seed

C. Green group-

- i Epidermis of seed green but embryo yellow
- ii Epidermis as well as embryo yellow

2. **Martain Classification-** This classification is based on shape and size of soybean seed

- i Soja elliptica
- ii Soja spherica ,
- iii Soja compressa

3. **Hertz Classification-** It is based on the shape of the pod

- i. Soja platycarpa
- ii Soja tumida

4. **American Classification-** This classification is based on maturity period. Varieties available in U.S.A. have been divided into ten groups according to their maturity period

Botanical Description-Soybean is an erect and bushy annual plant . The plant varies from 0.5 to 2.0 meter in height. The branches may be spreading or ascending, short or elongated. The root system of the soybean consist of a tap root , a large number of secondary roots and much branched adventitious roots. Tap root go as deep as 150 centimeter. The first nodules are visible about 10 days after sowing. The nodules are initiated in the soybean roots by *Rhizobium japonicum* as soon as root hairs are present on the root and at maturity the root system is extensively nodulated.

Leaves- The leaves of soybean are of four different types.

Cotyledons or seed leaves- These are elevated above the soil by rapid growth of the hypocotyls. These cotyledons become green and carry out some photosynthesis in addition to supplying stored minerals and reserve food to seedlings. Two simple, primary leaves - After the cotyledons have been elevated above the soil, the two primary leaves appear from the apical bud between them and within a few days are fully expanded . The primary leaves are oval in shape.

Trifoliolate leaves- All other leaves produced on the main stem and on the branches are trifoliolate and are alternately arranged. The leaflets are generally ovate and lanceolate in shape.

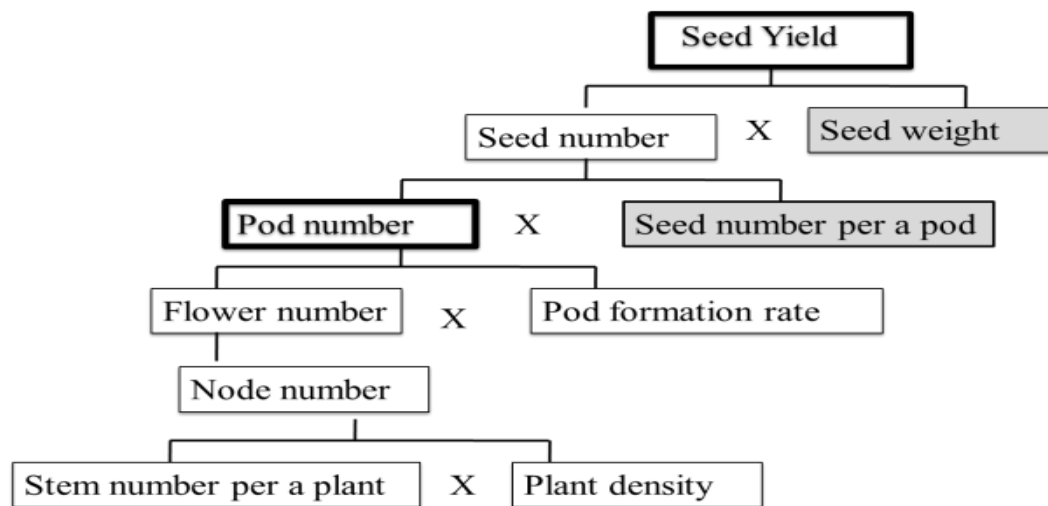
Prophylls- These are at the base of each lateral branch. These are tiny paired simple leaves, lack petioles and hardly reach a length of more than 1 millimeter.

Inflorescence – The flower are borne on short axillary or terminal racemes, and there are usually 8 to 16 flower in a cluster. The soybean flower is a typical papilionaceous and are normally self pollinated and completely self fertile.

Pods-The pods are small, either straight flattened or cylindrical. The number of pods varies from two to more than 20 in a single inflorescence and up to 400 on a plant. They usually contains 2 to 3 seeds pod

Seed-Seed vary in shape from spherical to somewhat flattened discs and in colour from pale green and yellow to dark brown . The seed coat is marked with a hilum or seed scar that varies in shape from linear to oval.

Yield Contribution-



Climate- Soybean grows well in warm and humid climate. A temperature of 26.5 °C to 30 °C appears to be the optimum for most of the varieties. In general it can be grown in tropical as well as subtropical regions receiving rainfall 750 to 800 mm

Soil- Medium to heavy, well drained soils with pH ranging from 6.0 to 7.5 are most suitable. Sodic and saline soils inhibit germination of seed. Salty and sandy, bad drained soils are not suitable. Water logging is injurious to the crop.

Land preparation- Soybean requires loose and friable seed bed hence one deep ploughing (15 to 20 cm) followed by clod crushing and 2 to 3 harrowing are sufficient

Seed and Sowing- The varieties of soybean are sensitive to photoperiod and require short day condition for flowering and therefore sowing time is important. The optimum sowing time for *Kharif* is First fortnight of June to 2nd week of July. The last week of January to first week of February for *summer* season.

Seed treatment- Seeds are treated with 1% mercurial compound or thiram or captan or Dithane M – 45 as a preventive measure against seed borne diseases. Trichoderma Viridae @ 5 gm/Kg preventive measure against soil borne diseases. Rhizobium culture :- inoculation of Rhizobium culture @ 250gm/10 Kg seed is recommended for enhancing activity of nitrifying bacteria for fixation of atmospheric nitrogen

- Methods of Sowing- Drilling
- Depth of Sowing- 4 to 5 cm
- Spacing :- For heavy soils – 45 X 10 cm, For medium soils – 30 X 10 cm

Seed rate- 75 Kg / ha

Fertilizers and manures- Application of 5 tones of FYM or Compost at last harrowing 50 Kg N/ha 70 Kg P₂O₅/ha at the time of sowing.

Intercultureing- Weeding and Hoeing- Weed free condition at initial stage is very essential. 2 weeding and 2 hoeing at 15 DAS and at 30DAS should be carried out

Chemical weed control-Pre emergence application of Nitrofen(Toke – 25) @ 1.5 Kg a.i./ha or Pre emergence application of Metribuzin (Sencor) @ 1.0Kg a.i./ha in 800 to 1000 liter of water is effective for controlling weeds.

Irrigation- Generally this crop is grown as a rain fed in Kharif if there is dry spell at flowering and pod formation, protective irrigation should be given at this stage. For summer cultivation- Total 5 to 6 irrigations should be given at an interval of 10-12 days.

Crop Rotation-Soybean – Wheat, Soybean – potato, Soybean – Gram, Soybean - potato – Wheat,

Harvesting and Threshing- At maturity leaves, stem and pods turn yellow also the defoliation of leaves started at this stage. The pods dry out quickly. The late harvesting leads to less yield due to pod cracking. The quality of the grain is spoil if harvesting is done early. The plants are cut and keep for drying 5 to 10 days in threshing yard and with the help of stick the grains are separated or threshing is done by using power thresher. After necessary drying the grains can be stored in gunny bags in cool and dry place.

Varieties-

Name	Duration (Days)	Av. Yield (q/ha)
MACS-58	90-100	25-35
MACS-124	90-100	25-35
JS-335	90-95	25-35
PK-1029	95-100	35-40
MACS-450	90-95	25-35
DS-228 (Phule Kalyani)	90-95	30-35

Insect pest-

1. Hairy Caterpillar- Adult lay eggs on lower surface of leaf in colonies. Larva feed on chlorophyll of leaves, resulting leaves look as veins only. **Control measure**- Manual collection and destruction of egg and larva masses. Dusting or spraying of any contact insecticide.

2. Girdle Beetle-After mating females make 2 parallel girdles on stem, female may bore several holes between these 2 girdles before inserting a single yellow egg inside the stem, resulting drying of plant. **Control measure**- Soil application of Forate 10 G @ 20 kg/ha.

3. Stem Fly- Serious pest in North India, Adult feed on leaves by making small multiple punctures, which appear as white spot on leaves. **Control measure**- Soil application of Forate 10 G @ 20 kg/ha.

4. Leaf Roller- Serious pest in M.P. and U.P. Female lay eggs either single or in groups on leaf surface. Small white larva fold leaves around themselves as a protection and feed inside, resulting papery skeleton of folded leaves. **Control measures**- Spray any contact insecticide at early incidence.

5. Leaf minor-The dark headed green or brown larvae mine in to tender leaves, which look like blister mine. The attacked plant do not grow properly. Insect remain active from July to December.

Control measures- Spray Metasystox or Dimecron.

6. White Fly-Serious pest in U.P and Delhi, Adults are small yellow bodies insects densely covered with waxy powder. They transmit yellow mosaic.**Control measure-** Spray Metsystox @ 0.1%.

Diseases-

Fungal-

1. Charcoal rot- Disease of roots and stem. Discoloured region becomes dark brown to black resulting death of infected plant.**Control measure-** Summer ploughing, crop rotation, balanced fertilizers, and seed treatment.

2. Rust- appears in high humidity condition in first week of September. Development of red pustules on leaves, usually on lower surface.**Control measures-** Destroy the affected plants, Spray Zineb for three times with 10 days interval.

13. Sesame, B.N. *Sesamum indicum* L., Family :- Pedaliaceae

Economic Importance- Sesame contains 50 % oil and 25% protein 100 g seed provides 592 calories Sesame oil is an important cooking oil in south India It is really the poor mans food substitute for ghee. Nearly 73% oil is used for edible purpose, 8.3 % for hydrogenation, 4.2 % for industrial purposes in the manufacture of paints, pharmaceuticals perfume, and insecticides. Lower grade oil is used in soap making industries, oil has medicinal uses. The oil cake is used as cattle feed .And as a manure. The oil cake contains 6.0 % N, 2.0% P and 1.0% K. India ranks first in area and production. Rs.1000 corer earn only from sesame export. 78% of the sesame seed produced in India is used for oil extraction. Only 2.5 % produce is used for planting purpose and rest of the seed is used in confection and in religious Hindu ceremonies.

Origin- According to Watt (1893) and Vavilov (1951) India considered to be the origin of sesame. While De Candolle (1886) opinion that sesame is originated in south-eastern Africa.

Classification- On the basis of Maturity period

- Early- varieties having less number of flowers and branches
- Late- varieties having more number of flowers and branches

On the basis of Seed colour

- White seeded- varieties having less number of flowers and branches
- Black seeded :- varieties having more number of flowers and branches

On the basis of Number of carpel's in the Capsule

- Bicarpellatum- having two carpel in the capsule
- Quadricarpellatum- having four carpel in the capsule

On the basis of Chromosome Numbers -

Group	Chromosome no.	Species
Gr. 1	2n= 26	<i>Sesamum indicum</i>
		<i>Sesamum alatum</i>
Gr. II	2n = 32	<i>Sesamum prostratum</i>
		<i>Sesamum lacintatum</i>
Gr III	2n = 64	<i>Sesamum radiatum</i>
		<i>Sesamum accidentale</i>

Botanical Description- It is a herbaceous annual plant growing to a height of 0.5 to 2.0 meters. In sesame two type of **root system** are observed, In early maturing varieties, the root system is poorly developed. The tap root bears only few secondary and tertiary roots. Late maturing varieties have a well developed root system in which tap root goes deeper and bears a large number of secondary and tertiary roots near the soil surface. The stem is erect, normally square in section. **Stem** colour can range from light green to almost purple, but the most common is darkish green shade. The **leaves** of sesame plant are very variable in shape and size. Generally the lower leaves tend to be broader while the upper ones narrow. The arrangement of the leaves on the stem may be opposite, alternate or mixed. **The inflorescence** is a raceme and the flowers arise in the axils of the leaves and on the upper portion of the stem and branches. The flower vary in size and colour. **The fruit** of sesame is a capsule having two or four carpel's. Numerous seeds are borne in the capsule. seed are small, ovate, slightly

flattened and somewhat thinner at the hilum. Colour of seed may be black, white, brown and their various shade.

Climate-

- Sesame is a warm region of the tropics & sub tropics.
- It require fairly hot temperature during its growth to produce maximum yield.
- A temperature of 25 °C to 27 °C encourages rapid germination, initial growth & flower formation.
- Low temperature at flowering is undesirable which results in premature flower drops.
- It is extremely susceptible to water logging & heavy rainfall Frost is also harmful to this crop.

Soil-

- Sesame can be grown on wide range of soils.
- The soil should be well drained.
- It grows well in sandy loam soil with adequate moisture.
- Very sandy, acidic or alkaline soils are not suitable.
- It can be grown in the soils having pH range of 5.5 to 8.0

Land preparation- The seed of sesame are very small and the tap root of sesame goes deep in soil an therefore it requires fine seed bed.Seed bed is prepared by ploughing the field and 2 – 3 harrowing and Planking.

Seed and Sowing-

Time of sowing-In North India it is cultivated in Kharif season only but in south India sesame is cultivated in both Kharif and Rabi season.Kharif sesame is sown immediately after on set of monsoon (15th June to15th July) the delayed sowing of sesame crop beyond 15th July decreases the yield considerably.

Seed Treatment-Seeds are treated with 1% mercurial compound or thiram or captan as a preventive measure against seed borne diseases like root rot, bud necrosis and wilt. Trichoderma Viridae @ 5 gm Kg preventive measure against soil borne diseases.Rhizobium culture- inoculation of Rhizobium culture @ 250gm/ 10 Kg seed is recommended for enhancing activity of bacteria for fixation of atmospheric nitrogen

Methods of Sowing- Drilling , Broadcasting

Depth of Sowing- 2 to 3 cm

Spacing - 30x 15 or 45 x 15 cm

Seed rate- 2. 5 to 3 Kg Drilling, 10 to 12 Kg/ha Broadcasting

Seeds being mixed with well sieved FYM or Fine sand for proper distribution

Manures and Fertilizers – Application of 5 tones of FYM or Compost at last harrowing or 1.0 tones of Castor cake at the time of sowing. 50 Kg N/ha (25 Kg N/ha at the time of sowing, 25 Kg N/ha at the 30 days after sowing)

Interculturing- Thinning- The first thinning was carried out at 10 days after sowing and second thinning was carried out at 20 days after sowing . Weeding - Weed free condition at initial stage is

very essential. 2 weeding and 2 hoeing at 20 DAS and at 30DAS should be carried out. Chemical weed control :- Pre planting application of fluchloralin or Pre sowing application of alachlor @ 1.5Kg a.i./ha in 800 to 1000 liter of water is effective for controlling weeds.

Harvesting and Threshing - At maturity leaves, stem and capsule turn yellow also the defoliation of leaves started at this stage. The capsule matures from base to upward. The plants are cut and tied in small bundles. They are dried and the seed is separated.

Varieties-

Name	Duration (days)	Av. Yield (q/ha)
No. 58-2	105-110	4.00
No. 128	120	2.50
No. 8	130	2.15
Phule Til No. 1	90-95	5.85
JLT-7 (Tapi)	76-83	7.00
JLT-26 (Padma)	75-80	6.50-8.50
JLT-408	80-85	7.50-8.00
Madhavi	75- 80	5.00-6.00
Punjab Til 1	80	4.00-6.00

Insect Pests-

1. Leaf Roller and Capsule Borer-In early stage of crop caterpillar feed on tender leaves, and remain inside the leaf web. At flowering the larvae feed inside the flower and at capsule formation, bore in to capsule and feed on the grains. **Control Measure**-Early sowing, crop rotation, prepare space for Bird stay in field, application of Phorate 10G @ 20kg/ha, Spray 0.05% Endosulfan.

2. Gall Fly- The maggots feed on flower bud, leading to formation of Gal like structure, instead of flower/capsule. **Control Measure**- Clipping of affected bud, Spraying of dimethoate 0.03%.

3. Hawk Moth-Caterpillar feed on leaves and defoliate the plant. **Control Measures**- Deep ploughing, expose the plant to Birds, Collection of caterpillars, dusting contact insecticide.

4. Bihari hairy Caterpillar-The caterpillar feed on leaves vigorously. **Control Measures**-Deep ploughing, expose the plant to Birds, Collection of caterpillars, dusting contact insecticide.

Diseases-

- 1. Phyllody**-All floral parts are transformed in to green leafy structure. The top portion of plant bend down. Such plant do not bear flowers and capsules.**Control measures**- delay sowing, intercropping with pigeon pea, remove affected plants, control vectors by soil application of Phorate 10 G @ 10 kg/ha or spraying dimethoate @0.03%
- 2. Phytophthora blight**-Initially water soaked spots appear on leaves and stem, the spots turn dark brown. The disease is sprayed fast in humid condition.**Control measures**- Deep ploughing, improve drainage, crop rotation, seed treatment with Trochoderma, Spray copper oxychloride.
- 3. Root and Stem rot**-Disease appear on roots and stem, the affected plant wilt off.**Control measures**- Deep ploughing, improve drainage, crop rotation, seed treatment with Trochoderma, soil drenching with copper oxychloride.
- 4.Cercospora leaf spot**-Disease appear as small angular leaf spot.**Control measures**- Early sowing, intercropping, treat the seed with Thiram, Spray Dithen M-45 @0.25%
- 5. Alternaria leaf spot**- Spots on leaves of brown circular or irregular in shape with concentric ring.**Control measures**- Use resistant varieties, intercrop with sunflower, spray Dithen M-45 @ 0.2%
- 6.Bacterial blight**-Water soaked small and irregular spots appears on leaves and increased in numbers and turned brown.**Control measures**- Seed treatment with hot water 52°C for 10 minutes, Steep the seed in Agrimycine 100 @ 250 ppm, foliar spray of Streptocyclin @ 500 ppm
- 7. Powdery mildew**- Small cottony spots appears on infected leaves, which gradually spread on lamina. Defoliation of plant.**Control measures**- Early sowing, resistant varieties, Spray wettable sulphur @ 0.2%

14. Niger (*Karala*), *B.N. Guizotia abyssinica* (L) Cass, Family- Asteraceae

Economic Importance :-

- Seed mostly used (75%) for extraction of Oil, as a food or confectioneries (18%) and as a bird feed (exported)
- Oil (35-40%), pale yellow colour, nutty test and pleasant odour, poor keeping quality due to high content of unsaturated fatty acids (Oleic acid 38% and linoleic acid 51.6 %).
- Oil is used for- 1. Culinary purpose.2. Anointing the body, 3. Manufacturing of paints and soft soaps and cosmetics.4. As a Lubricant, 5. As a base oil in Perfume (good absorbent)
- Cake- As a valuable cattle feed for milch animals, As a manure.
- Foliage-As a green manure.

Origin :- It is a native of tropical Africa (highlands of Ethiopia)

Area and Production in World-India has more than 50% of worlds area and production.In Ethiopia it is a major oil seed crop.Grown in Sudan, Uganda, Zaire, Tanzania, Malawi, Zimbabwe, Pakistan, west Indies, Nepal, Bhutan and Bangladesh as a minor oil seed crop.

Area and Production in India-Total area under Niger- 5.22 lakh ha.Total Production- 1.49 lakh ton and Productivity- 2.86 q/ha.

State	Area (000ha.)	Prod. (000 ton)	Av. Yield (q/ha)
Madhya Pradesh	202	43.4	2.15
Orissa	152	47	3.09
Maharashtra	67.4	18.9	2.80
Karnataka	44.0	8.0	1.82
Bihar	24.5	17.8	7.27
Andhra Pradesh	15.8	5.6	3.54

Botanical Description-Niger is dicot herb.**Roots-** Well developed with central tap root.**Stem-**Round, hollow, smooth to slightly rough and branched grow up to 1.5 m. height.**Leaves-**first leaf is paired and small and successive are larger and opposite. Leaves are 10-20 cm in length and 3-5 cm in width, softly hairy on both side. **Flowers-** Born in cluster at leaf axils. **Head** is like sunflower, comprising Disc and Ray flowers.Ray flowers are sterile, Disc florets those at the edge open first followed progressively by the next in line to the centre of the head as in case of sunflower. The anthers are yellow and stigma is hairy. The floret liberates pollen in morning. Insects are more important as a pollination agent.It is a cross pollinated crop. **Seed-** The seed is small 3-5 mm in length and 1.5 mm in width, lanceolate in shape.The testa is hard, glossy and usually black.Normally

1000 seed weight is 3-5 gm. There are about 15-30 seeds/head. Mostly the seeds at centre are immature.

Climate-Temperate region crop but can be adapted to semi- tropical climate at the altitude of 500 to 1000 MSL.Temp. range 18-23⁰C is favorable during growth period.Rainfall- 1000 to 1300 mm.800 mm but well distributed. At Flowering to maturity- heavy rains, high wind or hail affect the crop yield.

Soil-Can be grown on wide range of soils.Clay loam or Sandy clay are best.pH- 5.2 to 7.3. Tolerate waterlogged condition due to its ability to develop aerenchymas under these conditions.Heavy clay or Black cotton soils are not suitable for this crop.

Land preparation-Seed size is small, therefore leveled seed bed is required.One or two ploughing with two harrowing and leveling.

Time of Sowing-Kharif- Middle of June to early August.Rabi- Early September

Seed rate and Spacing-

- Method of sowing- Generally Broadcasted but line sowing by mixing with soil or ash is recommended.
- Seed rate for sole crop- 5-8 kg/ha.
- Spacing- 20 to 30 cm line sowing.
- Depth- 1-3 cm. according to moisture available.
- Under optimum moisture condition it will germinate with in 3-5 days.

Seed Treatment-Seeds are treated with 1% mercurial compound or thiram or captan or Dithane M – 45 as a preventive measure against seed borne diseases.Trichoderma Viridae @ 5 gm/ Kg preventive measure against soil borne diseases.

Fertilizers :-20 Kg N/ha . 50% N at the time of sowing&50% N at 30DAS.

Interculturing :-Thinning :- The thinning is very important in Niger. Thinning was carried out at 20 days after sowing. It recommended only 1-2 healthy seedlings at 10 cm distance. Niger crop should be kept weed free for first 35 days by giving 1 – 2 hoeing and 1 to 2 weeding depending upon weed intensity.

Insects Pests-Caterpillar, Semi lopper, Surface Grasshopper, Cut worm

Diseases-*Alternaria* leaf spot, *Cercospora* leaf spot, Powdery Mildew, Root rot

Harvesting and Threshing :-The crop matures in 4 to 5 months . It is cut closed to the ground and dried in sun for 4 to 5 days and threshed out by biting with sticks.

Yield :- 4 to 5 q /ha

Varieties-

Name	Duration in days	Oil %	Av. Yield (q/ha)
No. 5	90	37%	2.50
Ootacamund	115	39%	3.50
No. 12-3	115-120	41.5%	2.50
IGP-76	110-115	40.75%	3.50
Sahyadri	100-110	40.5%	5.00

15. Castor (Erandi), B.N. *Ricinus communis* L., Family :- Euphorbiaceae

Economic Importance- Castor is an important industrial oilseed crop. Castor oil is valuable on account of its non drying and non freezing quality at low temperature hence it finds a number of uses for domestic, medicinal, industrial purposes.

Uses of Caster oil - Castor oil contains hydroxyl fatty acid (85 %) known as ricinoleic acid. This ricinoleic acid can be treated as dehydrated oil, this dehydrated oil is in demand for paints and varnishes because of its non yellowing quality. Hydrogenated castor oil is used in polishes, ointment, waxes, printing ink, cosmetics, hair dressing, soaps and disinfectant. In dyeing industry it is used for the preparation of turkey red oil. In many veterinary uses. It is used externally as an emollient. Castor oil is the chief material for the production of sebacic acid which is the basic ingredient in the production of synthetic resins and fibers. As a lubricant in all moving parts of machinery. As a purgative, and laxative medicine.

Oil Cake- The castor oil cake is a valuable manure. The castor oil cake contains the poison substancericin, therefore it is unfit for cattle feed. The oil cake contains 5.5 % N, 1.8 to 1.9% P and 1.1% K . The cake has anti termite properties.

Plant- The beaten pulp from stem is a short fiber which is used in paper industry. Because of its deep root system, drought hardiness and quick growth, it finds a place of prestige in the cropping system of dry land agriculture in semi arid zones of India. Green leaves are used for rearing eri silk worms. It is a good shade crop grown in turmeric crop. As a wind breaker in Sugarcane and as a pest attractant in Tobacco.

Seeds- Effective control for store grain pest. A few castor seeds are mixed in safe storage of sesame.

Origin- According to De Candolle & Hooker castor is from tropical Africa. In Sanskrit literature it is said that castor originated in the tropical belt of both India and Africa.

Classification :- The *communis* species is grouped into six sub species

1. *Ricinus communis* Percicus
2. *Ricinus communis* Chinensis
3. *Ricinus communis* Zanzibarinus
4. *Ricinus communis* Sanguineus
5. *Ricinus communis* Africanus
6. *Ricinus communis* Mexicanus

Botanical Description- Roots- It has a Tap roots with well developed secondary roots, often deeply penetrating in soil to take maximum moisture. **Stem-** It is round and covered with waxy bloom which gives red or green stem a bluish appearance. There are well developed nodes on stem from which leaf arises. Castor is a multi branched crop. **Leaves-** Born on long stout petiole in alternate fashion. Young leaves are toxic to animals. The large leaves are dark glossy green. Temp. and water stress are important factors for growth of leaves. **Flower-** Inflorescence in form of pyramidal raceme also known

as spike or candal, born on terminaly on main and secondary branches. The lower portion of raceme bear male flower and the upper female. Pollen shedding time start before sunrise and extend up to 2 to 5 hrs. It is a cross pollinated crop and the wind and insects are the agents for fertilization. **Fruit/Capsule and Seed**-The fruit is a globular spiny capsule, which become hard when ripe. Each capsule contains 3 oval shaped seed.

Area and Production in world

Country	Area (000 ha)	Prod. (000 ton)	Av. Yield (q/ha)
India	690	842	12.20
China	225	889	3.95
Brazil	90	26	2.85
Ethiopia	15	16	10.7

Area and Production in India

State	Area (000 ha)	Prod. (000 ton)	Av. Yield (q/ha)
Gujarat	346	622	18.00
Andhra Pradesh	246	66.5	2.51
Rajasthan	79	39	5.04
Maharashtra	11	2.8	2.55

Climate-Castor is a drought resistant crop and grows well in relatively dry regions receiving about 500 to 700 mm rainfall. It requires moderately high temperature 20°C to 27°C with low humidity through out the growing season. Heavy rainfall at flowering reduces the yield. It is also susceptible to frost condition.

Soil-Castor can be grown on all type of soils. However, medium to light well drained soils are suitable. Heavy soils and marshy condition are unfavorable for proper growth. It can tolerate acidity (pH 5 to 6.5) the crop can not tolerate alkalinity of soil.

Land preparation-Due to its well developed and deep root system one deep ploughing (20 – 30 cm) followed clod crushing and By 2 to 3 harrowing are sufficient.

Seed and Sowing-

Time of sowing :- *Kharif* – IIIrd week of June to 15th July

Methods of sowing :- dibbling

Spacing :- a) 90 X 45 cm for Girija , V-9

b) 60 X 45 cm for Aruna

Seed rate 10 to 15 Kg /ha for Girija , VI-9

20 to 22 Kg /ha for Aruna

Depth of sowing :- 7 to 8 cm

Seed treatment-Seeds are treated with 1 g/Kg mercurial compound of thiram + 2.5 g/Kg Lindane as a preventive measure against seed borne fungus diseases i.e. twig blight, Alternaria blight, bacterial leaf spot. Trichoderma Viridae @ 5 gm/ Kg preventive measure against soil borne diseases.

Interculturing-The thinning is very important in castor, thinning was carried out at 15 days after sowing recommended with only one healthy plant at 45 cm distance. Castor crop should be kept weed free for first 60 days because it is widely spaced and slow growing crop. Weed problem is more hence 2 – 3 hoeing and 2-to 3 weeding are recommended depending upon weed intensity.

Chemical weed control-Pre emergence application of Diuron @ 1.1Kg a.i./ha or post emergence application of Dalapon @ 2.2 to 2.4Kg a.i./ha or Simazine @ 1to 1.5 Kg a.i./ha. or Eptan @ 3.4 Kg a.i./ha as a pre sowing spray is effective for controlling weeds. Nipping of Auxiliary buds-The auxiliary buds present on main shoot are nipped off soon after the emergence of primary spike . By keeping a single main spike which helps in reduction in life period of crop and increasing the yield.

Irrigation-The Castor is mainly a drought resistant crop. grown as a rain fed crop but soil should be having adequate moisture at the time sowing hence irrigation is needed as pre sowing irrigation and at flowering

Manures and Fertilizers-Generally castor is not manured but the neem cake if available, should applied @ 1.0 tones/h because it increases the oil %

Fertilizers :- 60Kg N+ 40 Kg P₂O₅ /ha
50% N + 100% P % at the time of sowing
50% N 40 to 50 DAS

Harvesting and Threshing-Castor crop matures in 145 to 280 days depending upon the varieties. Harvesting is done when the capsules turn yellow. However maturity of the capsule is not at the same time, therefore 2to 3 pickings may be carried out for harvesting of entire crop. The spike should be dried in the sunlight for 4 to 5 days and then threshed.

Yield :- Rain fed – 5 to 6 q/ha , Irrigated – 12 to 15 q/ha

Varieties-

Aruna 115-120	10-12 (Mah, A.P.)
Girija 115-120	13.5-14.5 (Mah)
G.A.U.C.H-1 100-110	16-18 (Mah)
G.C.H . -4 210-240	15-20 (Mah)
Saubhagya (A.P.), TMV-1 (TN), EB 16A (Bhiar)	
Type-2 (U.P.), CH-1 (Haryana), VI-9, J-44, Rosy	

Insect and Pest-Castor Semi lopper, Castor capsule and shoot borer, Jassids, Red Spider Mite,

Diseases-Seedling Blight, Twig Blight, Frog eye leaf spot, Alternaria Blight , Rust, Pod Rot Bacterial Blight,

16. Cotton ,B.N. *Gossypium spp.*, Family :- *Malvaceae*

Economic importance :-Cotton, the White Gold is one of the most important commercial fiber crops.It plays a prominent role in the national and international economy.Cotton is grown chiefly for its fiber used in the manufacture of cloth for the mankind.It is also used for several other purposes like making threads, for mixing in other fibers.The seeds content about 15-25 % of edible Oil. After extraction of oil from seed the cake is the rich source of organic matter which content about 6% N, 3% P and 2% K. Cotton seed, Cotton linters and pulp obtained during oil extraction and cotton meal are good concentrated feed for cattle

Origin- It is said to have two centers of origin one in the old world, viz. India, Indochina and tropical Africa, and other in the New world cotton viz. Mexico or Central America

Area and Production in World-Cotton is grown in more than 80 countries of the world, out of which India, USA, China, Pakistan, Turkey, Brazil, Turkmenistan, Greece, Australia, Uzbekistan, Argentina, and Egypt are major cotton producing countries, which accounts 82% of area and 86% of production of world.

Area and Production in India-India rank first in world in area and third in production.Cotton is cultivated on about 9 million ha area in 9 states of India.The total production was increased from 71 lakh bales (1977-78) to 171.2 lakh bales (1996-97)

Cotton growing Area in India-

- Northern region (hirsutum, arboreum)- Punjab, Haryana, North west Rajasthan, and western U.P.
- Central Region (herbaceum- arboreum- hirsutum)- Maharashtra, M.P., Gujarat and southern Rajasthan.
- Southern Region (hirsutum- arboreum- herbaceum)- Tamil Nadu, Andhra Pradesh and Karnataka.

Cotton area in Maharashtra-

- Vidarbha Tract- Area of 16 to 17 lakh ha and production of 8 to 8.5 lakh bales, Yawatmal, Akola, Amravati, Buldhana & Vardha districts , 85 to 90 % of the total area in the tract.
- Marathwada- Area of 7.5 to 8 lakh hectares and production of 2.5 to 3 lakh bales, Aurangabad, Parbhani, and Nanded districts, 80 to 85 % of the total area in the tract.
- Deccan tract- Area of 3.5 to 4.0 lakh hectares and production of 2.5 to 3.0 lakh bales. Rainfed cotton is predominant in Jalgaon, Dhule and part of Nasik districts , irrigated cotton is prevalent in the Deccan canal zone of Ahmednagar, Pune & Solapur districts.

Classification-*Gossypium* genus (true, lint bearing) consist 39 species, out of which 4 are cultivated.

Diploids

$$n = 13$$

Old world/Asiatic/Deshi cotton

Tetraploids

$$n = 26$$

New world/American cotton

G. arboreum
G. herbaceum

G. barbadense (Egyptian)
G. hirsutum (American)

In India

- Hybrids- 40%
- *G. hirsutum* 36%
- *G. arboreum* 16%
- *G. herbaceum* 8%

In USA

- * *G. hirsutum* 99%
- * *G. barbadense*

Botanical Description-It is a herbaceous plant, most of the cultivated cottons are annuals. Which attains height of 60 to 200 cm. The plant has a **tap root** with secondary roots. The main **stem** is erect and much branched, the branches develop from buds located at the nodes of the main stem. There are two buds at the base of each cotton leaf petiole. The true axillary bud develops in a vegetative branch, which only bears leaves and no flower. The accessory bud generally develops into fruiting branch. The **leaves** are spirally arranged on the main stem and the vegetative branches, except on fruiting branches where they form two alternate rows. The **flowers** develop at the node to a leaf in fruiting branches only. The flower buds which appear a small pyramidal shaped leafy structure are called as square. **Square** consists of three triangular shaped leafy structure known as bracteoles and the flower bud. The **flower** open about 18-24 days after the square are seen. The fruit is the enlarged ovary that develops into 3 to 5 loculed capsule or boll. **Bolls** vary in size and shape, but are usually more or less egg shaped. When the boll is ripe, the capsule cracks or splits along the lines or sutures where the carpel meet, and the cotton within expand greatly in a white fluffy mass. The number of seed in each boll may be 24-50. The cotton **fiber** is simply an elongation or outgrowth of an epidermal cell of the seed coat. The long outgrowth forms the staple or lint while shorter outgrowth form fuzz.

Climate- Cotton is a sub tropical crop, but it is cultivated successfully in many parts of the world away from the sub tropics. It can not tolerate continuous rains or long dry spell of particularly at flowering and fruiting stages as that may cause heavy shedding of buds and young bolls. Heavy humid or cloudy weather and poor light or sunshine conditions are equally harmful as they reduce the yield, promote the attack of pests and diseases and lower quality of the produce. Cotton plants Cannot stand frost. Optimum temperature range for the growth of the plant is 21°C to 27°C and for fruiting and boll development 27°C to 31°C during day followed by cool nights. A well distributed rainfall of 625 mm can give a good yield but a higher rainfall up to 1250 mm can be tolerated if cotton is grown on well drained soils. Heavy rains after germination retard plant growth, increase weeds and affect the crop yield adversely.

Soil- cotton is grown on a variety of soils viz. sandy soils of Rajasthan, rich alluvial loams of indo-gangetic plains, the fertile black cotton soils of Maharashtra and M.P. and red laterite soils of Tamil Nadu, A.P. and Karnataka. It was observed that alluvial soils are predominate in the Northern hirsutam- arboreum region. Black soils in the central arboreum region and black, red and yellow soil in the herbaceum region. In Eastern region, cotton is more or less restricted to red loams in Assam, Manipur and Orissa and red yellow in Tripura. Medium to deep black soils with good drainage is generally suitable for growing cotton in Maharashtra.

Preparation of land- If the land is under cotton in the previous year, a clean –up- campaign is very necessary. The debris of the previous year's crop along with diseased or pest attacked bolls fallen on the ground should be collected and burnt. This will help in reducing the attack from pest and diseases in early stages of plant growth. Cotton requires a fine and firm seedbed. For rain fed crop, a deep ploughing (up to 20 – 25 cm deep) followed by 2 to 3 harrowing will be sufficient, add 12 to 15 tones of FYM or Compost before last harrowing.

Rainfed Cotton or Dry cotton

Time of sowing- cotton is sown immediately after onset of monsoon (15th June to 15th July)

Seed Treatment- The seed may not get entangled into each other due to the fuzz. It is usually rubbed with mud or a paste of earth and fresh cow dung before sowing. Before doing this it would also be desirable to soak the Deshi as well as American seed in water for 2 to 3 hours and 4 to 6 hours respectively.

Seeds are treated with 1% mercurial compound (Carbendazime 1 gm or 3gm thiram or Captan as a preventive measure against seed borne diseases like wilt, and root rot. *Trichoderma Viridae* @ 5 gm / Kg preventive measure against soil borne diseases.

Methods of Sowing-Drilling, Dibbling

Depth of Sowing- 2 to 3 cm

Seed rate-

Desi cotton-	12 to 15 Kg /ha
American cotton-	5 to 6 Kg /ha
Hybrid cotton-	2.5 to 3 Kg /ha

Spacing-

Desi cotton-	45 X 22.5 cm by drilling
American cotton-	60 X 30 cm
Hybrid cotton-	60 X 60 cm, 75 X 75 cm, 90 X 90 cm, 120 X 90 cm by dibbling

Manures and Fertilizers- 12 to 15 tones of FYM/ Compost before last harrowing

Fertilizer-Deshi cotton-50 Kg N + 25 Kg P₂O₅ + 25 Kg K₂O/ha

American cotton:-80 Kg N + 40 Kg P₂O₅ + 40 Kg K₂O/ha

Hybrid cotton :-80 Kg N + 40 Kg P₂O₅ + 40 Kg K₂O/ha (50% N + 100% P & K at sowing &50% at 4 weeks after sowing)

Intercultural operations- Thinning and Gap filling should be completed within 2 – 3 weeks from sowing. 2 – 3 hoeing and 2 to 3 weeding should be done as per weed intensity. Chemical weed control- pre – emergence application of Butachlor or Fluchloralin @ 1.0Kg a.i./ha or Diuron @ 0.75 Kg a.i./ha is effective for controlling weeds.

Harvesting- Cotton is harvested in three or four pickings as the bolls mature. Kapas or seed cotton is removed from the bolls in the field itself and care is taken to see that no leaf portion gets attached to the fiber. Clean picked cotton fetches premium Price in the market. The harvesting period varies with the variety but generally extends over about 8 to 10 weeks. It starts in October and extends up to December in Maharashtra. The middle picking are the heaviest and best in quality. The last picking is usually of poor quality. The yield varies widely from region to region and from year to year depending on the agro climatic conditions.

Yield- Deshi cotton -6 to 8 q/ha, American and Hybrid Cotton – 10 to 12 q/ha

Varieties :-

1. **Deshi cotton-** Vidharba region – AKH-4, AKA-5, AKA- 8401, Y-1, Jyoti
Marathwada region – Eknath, Rohini, Namdev
Deccan tract :- Y-1, Jyoti and JLH-168

2. American varieties-

Vidharba region :- L-147, DHY-286, BURI-1007, SRT-1
Marathwada region:- SRT-1, Nagnath(PH-93), Purnima
Deccan tract :- SRT – 1, AKH-081

3. Hybrid varieties-

Vidharba region- PKV Hybrid-2, AHH-468, H-4, Varlaxmi, H-6, NHH-44,
Marathwada region:-NHH-44, NHB-12
Deccan tract :- H-4, H-6

Irrigated Cotton

Time of sowing :- Phaltan Baramati area – First fortnight of March, Ahmednagar area – First week of April, Deccan tract – Second week of May.

Seed Treatment- Same as in Rain fed crop.

Methods of Sowing- Dibbling

Depth of Sowing :- 2 to 3 cm.

Seed rate- American cotton- 4 to 5 Kg /ha, Hybrid cotton-2.5 to 3 Kg /ha

Spacing- Medium soil :- 90 X 90 cm

Heavy soil :- 120 X 90 cm

Black cotton soil :- 90 X 90 cm or 120 X 90 cm

Manures and Fertilizers- 12 to 15 tones of FYM/ Compost before last harrowing.

American cotton / Hybrid cotton :-100 Kg N + 50 Kg P₂O₅ + 50 Kg K₂O/ha (50% N + 100% P & K at sowing, 25% at one month after sowing & 25% at the time of flowering.)

Intercultural Operations- Same as in case of rain fed.

Water management-The water requirement of cotton is about 940 mm. Cotton plant is very sensitive both to deficient as well as excessive soil moisture conditions. It is therefore , very important to maintain adequate moisture in the soil at different growth stages of the crop. The water use by cotton plant is at the rate of about 2.5mm per day between emergence to square formation, 2.5 to 6.25 mm per day between square formation to first bloom and about 6.65 to 10.0 mm per day between first bloom to first open boll period.

Use of Drip Irrigation not only save the water but also maintain adequate moisture condition in soil, it also facilitate fertigation (application of liquid fertilizers) which Combine result in higher yield and profit.

Harvesting-Cotton is harvested in three or four pickings as the bolls mature. *Kapas* or seed cotton is removed from the bolls in the field it self and care is taken to see that no leaf portion get attached to the fiber. Clean picked cotton fetches premium Price in the market. The harvesting period varies with the variety but generally extends over about 8 to 10 weeks. It starts in October and extends upto December in Maharashtra. The middle picking are the heaviest and best in quality. The last picking is usually of poor quality.

Yield- The yield vary widely from region to region and from year to year depending the agro climatic conditions.

American cotton :- 16 to 18 q/ha, **Hybrid cotton** :- 20 to 25 q/ha

Varieties-

American Varieties :-Vidharba region- DHY-286, AKH-081

Marathwada region-MCU-5

Deccan tract- Laxmi, LRA-5166

Hybrid Varieties :-

Vidharba region- AKH-468,H-4, H-6, NHH-44

Marathwada region- NHH-44,H-4

Deccan tract- H-4, H-6, AKH-468

Quality-The economic product of cotton plant are- Seed and Lint.The quality depends up on- Variety, Environment and Management. Characters that's determine quality are-

Ginning percentage-ratio of lint to seed *baradense*-28-30%

hirsutums-34-38%

deshi- 36-42%

Colour-Mostly white. Egyptian cotton are cream colour, Green, Brown, Blue- under research

Trash-Foreign material like dry leaf, burs, broken seed etc. in the lint are known as trash. Present of trash deteriorates grade, processing efficiency and quality of yarn.

Fiber Quality-Physical properties viz- length, fineness, strength and maturity.

Fiber length-

Category	Staple length (mm)
Short	20.0 and above
Medium	20.5-24.5
Medium long	25.0-27.0
Long	27.5-32.0
Extra long	Above 32.5

Fiber fineness-Decide spinnable count of cotton and quality of yarn. The yarn spun from fine cotton will have large number of fiber in its cross section and will be stronger and uniform.

Based on micronair value fiber are classified as-

Very fine	< 3.0
Fine	3.0-3.9
Average	4.0-4.9
Coarse	5.0-5.9
Very coarse	> 6.0

Strength-Durability or wear life of textile material- expressed in term of tenacity defined as the load or force required to break a fiber of unit linear density. (g per tex or centi Neuton per tex). Pressley Strength Tester or Stelometer is widely used for determining the strength.

Spinning performance-Measured by number of counts. Count refers to number of hanks, each of 840 yards, which weigh to 1 lb.Up to 1950 cultivated cotton had only 38 counts.Today we have varieties with a count range from 6s to 120s.

Oil Content-This also decides the commercial importance.Generally seed content 14.6 to 25.6 % of oil.Presence of toxic pigment is a major problem.

Maturity- it indicates the degree of formation of cellulose in the fiber or thickening of its cell wall.In cotton there are three types of fiber viz. Thick walled or mature fiber, thin walled immature fibers and intermediate types called half mature fibers.

Cotton is tested for quality in the Technology Laboratory of the Indian Central Cotton Committee at Mumbai established in 1924 and also at the fiber testing station, Nanded and Achalpur.

Cropping Systems-North India- Wheat/Mustard – Cotton,

Dry land of Central India- Sorghum / Soybean – Cotton

Intercropping- Soybean, Black gram, Green gram

Insect and Pest-

1. Pink boll worm-It is a notorious pest of cotton. It affects the cotton yield and quality, by sealing the tips of flower petals, resulting rosette flower, cut through the developing lint to the seed resulting retarded lint development, it eat away the internal portion of seed as it is its preferred food.

2. Spotted boll worm-The incidence is noticed at early stage of crop. The larva enters the terminal bud of shoot and tunnels downwards. The shoot withers and dies. Later the larva bore in to developing flower bud and feed on the pollen causing the condition known as ‘flared’ squares. As the boll develop, the larva bore in to the unripe boll, the boll turn brown, rot and drop off.

3. American boll worm- Commonly known as Gram Pod Borer, it causes damage to cotton crop by boring in to bolls, also feed on tender top shoot, flower bud and bolls.**Control Measures-**Integrated Pest Management- Clean cultivation, seed treatment (heating seed at 60⁰C for few min), or fumigation with methyl bromide, Use of Pheromone Trap, Spray Neem seed extract and Heliokill at early stage. Use BT Cotton varieties. Pathogen used- *Bacillus thuringiensis*

4. Jassids, Aphids, Thrips, and White fly, Cotton leaf roller, Mealy bug.

Small insects suck the sap from leaves, flower buds and developing boll. The infected plant turn pale yellow, wither and die. Spray the crop with Dimecron 100 EC @ 250 ml per ha. Or Metasystox 25 EC @ 1 lit /ha.

Diseases-

1. Root rot-Caused by soil born fungus, it appears in patches, healthy plant may wilt in 24 hrs., resulting in low plant stand and low yield. Seed treatment, Adopt crop rotation.

2. Fusarium wilt- Soil born disease, stunted plant growth followed by yellowing, wilting and dropping of most of leaves. Seed treatment, Use healthy seed, apply sufficient quantity of potash and organic matter.

3. Anthracnose- It is a seed born, dark brown lesions on plant below leaf surface, circular water soaked spots on leaves and bolls. Seed treatment, Spray copper fungicides @ 2 kg/ha.

4. Bacterial blight-Seed born, infected by plant residues, Same type of spots appear on all plant part, resulting in drop off of leaves, Squares and bolls. Seed treatment, clean cultivation, Spray copper fungicide.

17. Jute, *B.N. Corchorus capsularis*, *Corchorus olitorius*, Family :- *Tiliaceae*

Economic Importance-

- Next to Cotton, Jute is an important fiber crop.
- Fiber of about Rs.200 crores is exported annually from India hence it is called as “the golden fiber”.
- The fiber is used for preparation of gunny bags, carpets, rugs, ropes, twins, nets and rough cloth.
- The stalks are used as fuel and also for making gun powder charcoal.
- The paper industry has been using it as a raw material for coarse paper.

Origin- The species *Corchorus capsularis* mostly found in Indo-Myanmar region, however the species *Corchorus olitorius* mostly found in Africa.

Area and Production in world-

- Grown on 19.29 lakh ha. area with average production of 32.67 lakh ton in India, Bangladesh, China, Thailand, Indonesia, and Nepal.
- India and Bangladesh together produce over 80% of the world production.
- In India it is grown on about 9.58 lakh ha. area with average production of 16.61 ton, in West Bengal, Assam, Bihar, Orissa, Tripura and Uttar Pradesh.
- In Maharashtra jute is not cultivated for fiber.

Classification- All the cultivated varieties are classified into two types of cultivated species.

1. *Corchorus capsularis* It is hardy in nature and can grow well both on high and low lands and is able to tolerate water logging conditions to some extent. It is more popular and nearly 70 % of total area is under this species.

2. *Corchorus olitorius* – It is grown on well drained high lands only

Botanical Description- It is herbaceous annual. The plant has tap root system with numerous lateral branches.

Plant character	<i>Corchorus capsularis</i>	<i>Corchorus olitorius</i>
Plant height	Plant is shorter in height 1.5 to 4 m	Plant are generally taller in height 1.5 to 5 m
Leaves	Glabrous, usually 5 to 13cm X 2.5 to 8.0cm , ovate oblong, coarsely toothed . Bitter in taste due to presence of Corchorin a bitter substance known as tita pat	Glabrous, usually 7 to 18cm X 4 to 8.0cm , oblong, coarsely toothed almost tasteless known as Mitha pat
Flowers	Small. 0.3 to 0.5cm in length, yellow in colour	Big, about 1cm long in length, yellow in colour

Seed pod	Globular or pear shaped, 1 to 1.5 cm in diameter, wrinkled, rarely smooth and 5 loculed	Long and cylindrical, 6 to 10 cm long, 0.3 to 0.8 cm in diameter, ridged lengthwise and 5 to 6 loculed
Seed	Seeds small, chocolate brown in colour and oval in shape. About 35 to 50 capsules. Weight about 300 seed/ gram	Pyramidal in shape and bluish green to steel gray or black in colour. Weight about 500 seed/ gram
Fiber	Fiber is whitish in colour and is known as white jute in commercial trade	Fibre is finer, softer, stronger and more lustrous. Yellow to gray or even reddish in colour and is known as tossa in commercial trade

Climate-

- Jute thrives best under warm and humid climate with temperature ranging from 24 to 37°C
- Well distributed total rainfall of 1500 mm, of which 250 mm precipitation from March to May is the most suitable for jute cultivation.
- Alternate period of sunshine and rainfall are beneficial.
- Young Jute plants are very sensitive to water logging condition.

Soil-

- Jute can be grown on all kinds of soils from clay to sandy loam but loamy alluvial soil are most suitable.
- Laterite and gravelly soils are not suitable for jute crop.
- Jute thrives best in the normal soil reaction i.e. pH 6 to 7.5 Below or above this pH, yield of crop suffers.
- Sandy soils and heavy clays are unsuitable for jute cultivation.

Preparation of land-

- Jute requires a clean, clod free and fine tilth for successful establishment.
- A deep ploughing (up to 15 – 20 cm deep) followed by 2 to 3 harrowing and one planking will be sufficient to prepare the fine seedbed.

Seed and sowing-

Seed treatment-Seeds are treated with organo mercurial compound of Agrosan or Ceresan @5 g/Kg as a preventive measure against seed borne diseases.

Sowing time-*Corchorus capsularis* (White jute) March – April . *Corchorus olitorius* (Tossa Jute) April – May

Methods of Sowing :-Drilling

Depth of Sowing :- 2 to 3 cm.

Spacing- White jute 30 X 10 cm, Tossa Jute 20X 10 cm

Seed rate- White jute 6 to 8 Kg /ha, Tossa Jute 4 to 6 Kg /ha

Manures and Fertilizers- 5 to 8 tones of FYM/ Compost before last harrowing.

Fertilizers- White jute :-60 Kg N + 30 Kg P₂O₅ + 30 Kg K₂O/ha

Tossa Jute :-60 Kg N + 30 Kg P₂O₅ + 30 Kg K₂O/ha

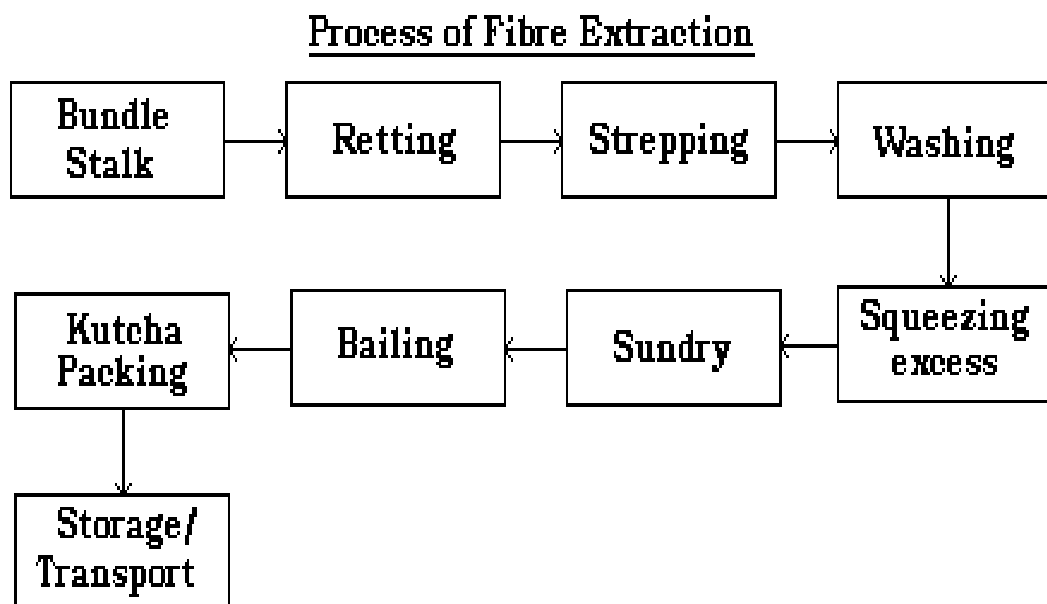
(50% N + 100% P & K at sowing & 50% at 4 to 5 weeks after sowing)

Intercultural operations-Thinning at 2 to 3 weeks after sowing, and at 5 to 6 weeks after sowing. hoeing and weeding should be carried out within 5 to 6 week after sowing.

Harvesting- The Jute crop is harvested for fiber at any time before flowering between 120 and 150 days after sowing. Early harvesting gives low yield but fine quality fiber, whereas late harvesting gives high yield but a coarse quality fiber. The ideal stage for harvesting is when the plants are in small pod stage, 135 – 140 days after sowing. Harvesting is done by cutting the plants close to ground level with sharp sickle. The harvested plants are tied in bundles and left in the field for 2 to 3 days for shedding of leaves. The apical portion of the plant may also be removed and left in the field.

Retting-

- Retting is a process by which the fibers in the bark get loosened and separated from the woody stalk. It is a microbial process affected by various aerobic and anaerobic micro-flora.
- Retting is best done in a shallow canal with slow running clear water, where such facilities are not available, tank with a depth of about two meters are quite suitable.
- The harvested jute bundles should be kept in 30 –60 cm deep water for 3 to 4 days before the entire bundle is steeped. Later on bundles are arranged in two to three layers and tied together. They are covered with aquatic weeds.
- The float is then weighed down with concrete blocks or kept submerged at least 10 cm below the surface of water.
- Retting is best done at 34 °C. At the end of the 8 day onward, the rets (stem) are to be examined. If fiber slips out easily from the wood on pressure from the thumb and fingers, retting is considered complete. Over retting results in dazed weak fiber.



Yield-*Corchorus capsularis* (White jute) fiber- 10 to 11 q/ha, seed 3.5 to 4.5 q/ha

Corchorus olitorius (Tossa Jute)- fiber- 15 to 16 q/ha, seed 1.5 to 3 q/ha

Varieties :-

Corchorus capsularis (White jute)- JRC-206, JRC-212, JRC 412, JRC312 (SONALI), EC-4142, EC- 4143, SHYMALI (JRC-7447).

Corchorus olitorius (Tossa Jute)-JRO 620, JRO 753, Baishaki tissa (JRO – 632) , Chaitali tisa (JRO-878), Navin (JRO 524) .

Insect pest- Yellow mite, Jute stem weevil, Jute semi lopper and Bihar hair caterpillar. **Control measures-** use of resistance varieties, proper crop rotation, late sowing for yellow mite, early sowing for semi lopper, spray of contact insecticide.

Diseases- Seedling blight, collar rot, root rot, Anthracnose are common. **Control-measures-** liming of acid soil, proper crop rotation, proper drainage, adequate potash, seed treatment, spray of Dithane-M- 45, copper oxychloride, carbendanzim.

18. SUNHEMP, B.N. *Crotalaria juncea* L. Family :-*Leguminosae*

Economic Importance -

- It plays an important role in the national economy both as raw material for indigenous industry and also as a foreign exchange earner through export.
- Recently it has gained additional importance because of increasing demand of a particular grade of fiber (White) for manufacture of tissue paper and paper for currency as it contains high % of cellulose and low amount of lignin.
- Fiber is used for making ropes, twines, net sand, rough cloths, gunny bags and stands next to jute.
- It is grown for green manuring purpose when incorporated in soil It adds about 40-60 Kg Nitrogen / ha

Origin -Its centers of origin is India.

Area and Production in World-It is grown in U.S.S.R., Romania, India, China, Hungary, Poland, Turkey, Brazil, Chili and Bangladesh.

Area and Production in India-In India it is cultivated on 1.4 lakh ha. With a production of 50,000 tons of fiber. It is grown in U.P., M.P. Maharashtra, Rajasthan and Orissa.

Botanical Description-It is an erect annual with few lateral branches growing up to 1.2 to 2 meter height. The stem is straight and thin. With very thin outer skin which contain long fibers. The plant is deep rooted with well developed lateral roots, all covered with numerous nodules. The leaves are almost sessile, small and narrow, shining and covered with short silky hairs. The inflorescence is a raceme, usually of 12-20 bright yellow flowers. Flowers are self sterile and are cross fertilized. The pods are small, cylindrical and about 5 cm in length. The seeds when ripe rattle with in the pod when shaken, from which the name *Crotalaria* has been derived. The seeds are of kidney shape.

Climate –Sun hemp is a warm season crop and grows best in tropical and sub tropical region of the world. In northern India it is grown in Kharif season, while in the south India it is cultivated in Rabi season. It requires a minimum of 400 mm of well distributed rainfall during the growing season for its successful cultivation for fiber purpose.

Soil – Well drained loamy soils are best for sun hemp cultivation. If there is proper management for drainage it can also be grown on heavy soils. Acid soils are not suitable for this crop. When grown for fiber it is sown on heavy waterlogged soils but for green manuring purpose it can be sown on heavy soils too.

Preparation of land -Sun hemp requires a clean and friable seedbed. A deep ploughing (up to 15 – 20 cm deep) followed by 2 to 3 harrowing, the field should be leveled by giving a gentle slope to ease drainage. There should be proper moisture in the soil at the time of sowing for proper germination of seeds.

Seed treatment :-Seeds are treated with 1% organo mercurial compound of Thiram 3 g/Kg as a preventive measure against seed borne diseases.

Sowing time :- The optimum sowing time for Kharif 15th June – 15th July. *Rabi- September – October*

Methods of Sowing :-Drilling for fiber purpose. Broadcasting for Green manuring purpose

Depth of Sowing :- 2 to 4 cm

Spacing :- 30 X 10 cm

Seed rate-Drilling :- 20 to 25 Kg /ha

Broadcasting - 55-60 Kg / ha

Fertilizers - 25 Kg N/ha + 50 Kg P₂O₅ /ha at the time of sowing

Interculturing -Weeding and Hoeing :- Weed free condition at 25 to 30 days is very essential. 1 or 2 weeding and 2 hoeing at should be carried out.

Harvesting –The harvesting of sun hemp for fiber extraction should be done at the pod formation stage for good quality fiber. Such fibers have a good colour and luster. Some times the plants are allowed to remain in the field until they are dead ripe, the fibre obtained from such plants is of very poor quality. Too early as well as too late harvesting spoil the fibre quality.

When grown for green manuring, it becomes ready for incorporation in the soil within 2 months after sowing (at flowering). At this time, the stalk will be tender with very little fiber formation with high nutrients and will take less time for decomposition in the soil.

Harvesting of fiber crop is done by cutting the plants close to the ground with sickle. The leafy top portion of the plants may be chopped off and used either as fodder or may be ploughed down to add organic matter to the soil. After 2 to 3 days when most of the leaves get dried up, these plants are shaken to shed the leaves. The plants are then tied into bundles of convenient sizes having 50 to 100 plants in each.

Steeping, Retting and Extraction-

The bundles are brought to the nearest ponds, ditches, pools or stream or shallow canal or river with slow running water and arranged side by side to form a platform in water for steeping. Then the bundles are covered with aquatic weeds like water hyacinth, grass etc. and weighed down 10 –15 cm below the water by stone or concrete blocks. Retting in slow running water is better than in stagnant water. This process is complete when the fiber is loose enough for extraction and easily separated from the sticks. This can be determined by examining 1 or 2 plants from the bundle after 2 to 3 days of retting.

The period of retting may vary from 3 to 15 days depending upon the temperature of retting water and month of harvesting. The optimum temperature for retting is 21 to 27 °C. The extraction of the fiber of sun hemp is more difficult than jute. Therefore, fiber is extracted single plant wise by breaking the lower end of the plant and then stripping upward from the bottom.

After extraction, the peeled fiber should be washed thoroughly in clear water to remove the dirt and other adhering materials. The fibers need to be squeezed to remove excess water and spread on bamboo rafts to dry in mild sunlight for 2 to 3 days. After drying the fiber is graded and bundled for marketing.

Yield- Fiber purpose – 8 to 10 q/ha

Diseases- Wilt, Anthracnose, Rust,

Insect and pest- Stem borer, Green bug

Varieties :-

1. K-12 black	6 to 10 q. fiber / ha
2. K-12 Yellow	14 q. fiber / ha
3. Nalanda sannu	6 to 10 q. fiber / ha
4. M-19	10 to 12 q. fiber / ha
5. M-35	10 to 12 q. fiber / ha
6. D-IX	8 to 12 q. fiber / ha
7. ST-55	10 to 12 q. fiber / ha

Dhaincha, B.N. *Sesbania aculeata*(Linn) Pers., Family :- Fabaceae

Economic Importance :-

- Dhaincha is a fast growing leguminous crop used for green manuring.
- It is capable to add 80 Kg N /ha
- In short duration (45 days) it grows up its maximum.
- It adds 20 –25 ton/ha. of organic matter in soil.
- In waterlogged soils it grows upto 1.5 to 1.8 meter high
- It has capacity to withstand in 60 cm high water

Area of production- Dhaincha is grown as green manure in Andhra Pradesh, Karnataka, Kerala, Tamil Nadu, Maharashtra, Bihar, U. P., M.P., West Bengal, Haryana, Rajasthan.

Climate- Dhaincha can be grown in different climatic conditions including adverse situations. However it grows well in warm and humid climate.

Soil- Dhaincha can be grown in different types of soil. It can also be grown on saline soils.

Land preparation- It does not require fine seedbed hence there is no need of ploughing only two harrowing are sufficient. Planking is done after each harrowing, and the land is leveled before sowing.

Seed treatment-Seeds are treated with 1% organo mercurial compound or Thiram 3 g/Kg as a preventive measure against seed borne diseases

Sowing time- When it is grown as a rain fed crop, sowing can be done in July – August. The sowing time in other season depend up on availability of water. If irrigation facilities are available it may be sown in May.

- Methods of Sowing :-Drilling and Broadcasting.
- Depth of Sowing :- 2 to 4 cm
- Spacing :- 40 – 50 X 3.5 to 4 cm
- Seed rate :- Drilling :- 30 Kg /ha
- Broadcasting :- 60 Kg / ha

Fertilizers- 25 Kg N/ha + 50 Kg P₂O₅ /ha at the time of sowing.

- Soil Improvement- If the pH of the soil is more than 10 Then add 75 – 80 quintals Gypsum/ha before sowing.
- The field is filled with water for 8 – 10 days, so that leaching process will be completed.
- Afterwards this, the water is drained out, and Dhaincha is sown in the field.

Interculturing-No need of intercultural operation, when grown for green manuring. But for seed production weeding and hoeing is necessary.

Irrigation- Summer sown crop is irrigated at an interval of 1 week up to 45 days. A total of 5-7 irrigations are required for the whole crop duration, where as irrigation of Kharif season crop depends upon the rainfall.

Incorporation of crop- When the crop is in flowering stage(40-45 days after sowing) it is incorporated in the soil with the help of M.B. Plough, manure trampler or disc harrow.

Crop Rotation- If irrigation facilities are available , the crop may be sown in May. After incorporating into the soil, transplanting of Rice is done during Kharif season. If dhaincha is sown in Kharif season , then in Rabi season crop like Wheat, Potato, Barley or Oat can be grown.

Harvesting and Yield- Seed crop of dhaincha is sown in July and harvested either in the end of October or beginning of November, 10 – 15 q seed /ha is obtained. When grown as green manure crop it gives about 20 to 25 tons/ha of green matter, which incorporated in soil and about 80-115 Kg of N/ha is stored in the soil.

19. Forage crops.

Importance-

- To provide nutritious food to our milk as well as draft animals, both the Cereals and Pulses are included in forage crops.
- The cereals are always high yielding, produce high green fodder and contain about 9-10% Protein, minerals and crude fibers.
- The pulse fodder is comparatively giving lower yield but high Protein, Carbohydrates and mineral.
- Therefore combination of both for feeding live stock is recommended.
- The forage crops can be stored by preparing Silage.

Forage Crops- Cereals- Sorghum, Pearl millet, Maize, All grasses.

Pulses-Cowpea, Cluster bean, Berseem, Lucerne

1. Name of Crop	Sorghum	Pearl millet	Maize
2. Botanical Name	<i>Sorghum bicolor</i> L.	<i>Pennisetum glaucum</i> L.	<i>Zea mays</i> L.
3. Soil	Medium to heavy, well drained	Medium to heavy, well drained	Medium to heavy, well drained
4. Varieties	Ruchira, Phule Amruta, M-35-1, SSG- 59-3. MP Chari and Pusa Chari.	Joint Bajara and Rajco Bajar	African Tall, Manjari comp., Vijay and Ganga Safed.
Name of Crop	Sorghum	Pearl millet	Maize
5. Sowing time	June-July, Sept.- Oct, Feb-March.	June-July, and Feb-March.	June-July, Oct- Nov, Feb-March.
6. Method of Sowing	Drilling at 30 cm.	Drilling at 30 cm.	Drilling at 30 cm.
7. Seed rate	40 kg/ha	10 kg/ha	75 kg/ha
8. Manures and fertilizers	5 tons/ha of FYM + 100:50:40 kg. NPK/ha	5 tons/ha of FYM + 60:30:30 kg. NPK/ha	5 tons/ha of FYM + 100:50:50 kg. NPK/ha
9. Harvesting	At 50% flowering	At 50% flowering	At Tassel initiation
10. Av. yield	50-55 tons/ha	45-50 tons/ha.	50-60 tons/ha.

Grasses

1. Name	Pavana grass	Hy. Napier grass	Rhodes grass
2. B. Name	<i>Sehima nervosum</i>	<i>Pennisetum purpureum</i>	<i>Chloris gayana</i>
3. Varieties	Motha Pavana -100-5	Yashwant (RBN-9), Phule Jaywant (RBN-13)	Rhodes-10, Rhodes-22-7-10
4. Sowing time	June-July, Feb- March	June-July, Feb- March	June-July, Feb- March
5. Spacing	90x60 cm	90x60 cm	90x60 cm
Name	Pavana grass	Hy. Napier grass	Rhodes grass
6. Seed rate	40,000 root stalks/ha	40,000 root stalks/ha	40,000 root stalks/ha
7. Manures and fertilizers	10 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting	10 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting	10 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting
8. Irrigation	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer
9. Intercultural operations	Weeding after each cutting	Weeding after each cutting	Weeding after each cutting
10. Harvesting	First cutting at 2-3 months after planting, then after at 1-1.5 months	First cutting at 2-3 months after planting, then after at 1-1.5 months	First cutting at 2-3 months after planting, then after at 1-1.5 months
11. Av. Yield	10-15 tons/ha	20-25 tons/ha	10-20 tons/ah

Name	Guinea grass	Para grass	Marvel grass
B. Name	<i>Panicum maximum</i>	<i>Brachiaria mutica</i>	<i>Dichanthium annulatum</i>
Varieties	Hamil, PGG-1	It is an Aquatic weed originated from Africa	Marvel-7, Marvel-8, Marvel-40, Marvel-93
Planting time	June-July, Feb- March	June-July, Feb- March	June-July, Feb- March
Spacing	60x60 cm	60x60 cm	6x30 cm
Seed rate	50,000 root stalks/ha	50,000 root stalks/ha	1 lakh root stalks/ha
Manures and fertilizers	10 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting	10 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting	10 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting
Irrigation	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer
Intercultural operations	Weeding after each cutting	Weeding after each cutting	Weeding after each cutting
Av. Yield	10-15 tons/ha	10-15 tons/ha	10-15 tons/ah

Name	Dinanath grass	Blue Panic grass	Anjan grass
B. Name	<i>Pennisetum pedicellatum</i>	<i>Panicum antidotale</i>	<i>Cenchrus ciliaris</i>
Varieties	PS-3, IGFRI-4-2-1, IGFRI-43-1, T-15, PP-33	-	Kajari-75
Planting time	June-July, Feb- March	June-July, Feb- March	June-July, Feb- March
Spacing	60x60 cm	60x60 cm	6x60 cm
Seed rate	50,000 root stalks/ha	50,000 root stalks/ha	50,000 root stalks/ha
Manures and fertilizers	5 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting	5 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting	5 tons/ha of FYM + 50:40:20 kg. NPK/ha+25 kg. N/ha after each cutting
Irrigation	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer
Intercultural operations	Weeding after each cutting	Weeding after each cutting	Weeding after each cutting
Av. Yield	10-15 tons/ha	10-15 tons/ha	10-15 tons/ah

Name	Cowpea	Cluster bean	Berseem	Lucerne
B. Name	<i>Vigna unguiculata</i>	<i>Cyamopsis tetragonaloba</i>	<i>Trifolium alexandricum</i>	<i>Medicago sativa</i>
Varieties	Sweta, EC-4216, Bundel Lobia, UPC-5286	Pusa Sadabahar, Sirsa-1, Sirsa-2, Guar-80, HFG-119	Miscavi, Berseem Ludhiana, Pusa Giant	Sirsa-8, Sirsa-9, Co-1, Moopa, Rambler, Anand-2, RL-88
Planting time	June-July, Feb-March	June-July, Feb-March	June-July, Feb-March	Sept.- Oct
Spacing	Broadcasting or 30 cm. line soing	45x15 cm.	Broadcasting	Broadcasting or 20 cm. line sowing 45-60 cm on Ridges and furrow
Seed rate(kg/ha)	35-45	40-45	25-30	20-25 Brct 12-15 line
Manures and fertilizers	5 tons/ha of FYM + 25:50:25 kg. NPK/ha	5 tons/ha of FYM + 20:40:0 kg. NPK/ha	15 tons/ha of FYM + 25:50:0 kg. NPK/ha	5 tons/ha of FYM + 20:60:0 kg. NPK/ha
Seed treatment	<i>Rhizobium</i> 250gm/kg	<i>Rhizobium</i> 250gm/kg	<i>Rhizobium</i> 250gm/kg	<i>Rhizobium</i> 250gm/kg
Irrigation	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer	At interval of 10-15 days in rabi and summer
Name	Cowpea	Cluster bean	Berseem	Lucerne
Intercultural operations	Weeding at early stage	Weeding at early stage	Weeding at early stage	Weeding at early stage
Insect pest	Hairy caterpillar, leaf hoppers, cut warms	Bihar caterpillar, Jassids.	Black ants, Surface grass hoppers.	Lucerne weevil, Caterpillar, Leaf hoppers.
Av. Yield	5-6 tons/ha	2.5-3 tons/ha	8-11 tons/ah	8-10 tons/ah

TARUN MEENA